flyPad performance calculators

Takeoff (A380 new, A320 with MCDU uplink), landing and descent calculators on the flyPad, the company datalink reply time of the A380X FMS, and the later additions: A380 DERATED takeoff, BTV braking mode, 40 kt landing crosswind, descent target waypoint and T/D slider, and shared landing calculators that keep calculating above the tailwind limit

Aircraft	A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only)
Where to find it	flyPad > **Performance** (Takeoff, Landing, Top of Descent tabs); flyPad > Settings > Realism; A380X MFD PERF T.O / INIT / WIND; A32NX MCDU PERF TAKE OFF
Branch	`core/efb-performance`, based on `core/a380x-mfd-fms` (stacked: needs `core/surveillance` and `core/a380x-mfd-fms` merged first). `core/efb-ground` is stacked on it (the taxi page uses its runway exits).
Status	Landing calculator (both aircraft) with the BTV braking mode and the 40 kt crosswind, descent calculator (A32NX, A380X), descent target waypoint and the fix for an FMS without waypoints: tested in the sim by the user, OK. A380 takeoff calculator: used in the sim with the company T.O flow, TOW in tonnes and CG envelope edges OK; its DERATED takeoff (D04-D24): tested in the sim by the user, OK (flyPad to MFD and INSERT, E/WD level and N1, FMA MAN DTO, ENG THR LEVERS NOT SET at TOGA). Company datalink reply time: tested in the sim; the company wind part (reply time kept after the WIND page merge, one request per flight plan, NO COMPANY REPLY after 4 min) is covered by the WIND page sim test (PASS). A320 takeoff calculator: flyPad part used in the sim, MCDU uplink pages checked in the browser harness only. T/D slider: no separate sim record. All calculators have unit tests (back-filled and mutation-tested). Shared landing calculators above the tailwind limit (placed 2026-10-18): not sim-tested on the core branch; calculator unit tests.
Pictures	flyPad, MFD and MCDU pages in the local browser harness, running the build installed in the sim (from the feature manuals of September 2026). The harness has no navigation database and no FMS: runways and inputs are typed, uplinks simulated. The DERATED takeoff pictures come from the same harness (current flyPad look). The other later additions (BTV mode, 40 kt, target waypoint) have no pictures.

1. A380 takeoff calculator and company T.O data
2. A380 takeoff calculator: DERATED takeoff
3. A320 takeoff calculator and MCDU uplink takeoff data
4. A380X company datalink reply time
5. Landing calculator
6. A380 landing calculator: BTV braking mode
7. A380 landing calculator: 40 kt crosswind on dry and wet runways
8. Descent calculator
9. Descent calculator: T/D distance slider
10. Descent calculator: target waypoint from the flight plan
11. Descent calculator with an FMS that sends no waypoints
12. Shared landing calculators above the tailwind limit

1. A380 takeoff calculator and company T.O data

> **Aircraft:** A380X (the flyPad screen is shared with the A32NX, next feature) | **Where:** flyPad > Performance > Takeoff; MFD PERF T.O | **Status:** Used in the sim by the user with the company T.O data flow; TOW in tonnes and the CG envelope edges tested OK. Unit tests against the Airbus charts and the FCOM.

What it is for

Before each takeoff the crew needs the takeoff weight limit of the runway, the FLEX temperature (reduced thrust), the speeds V1, VR, V2 and the flaps. On the real A380 they come from the OIS takeoff application or from the airline (company takeoff data, sent to the FMS by datalink). The flyPad calculator does the computation; SEND TO FMS plays the airline: the MFD receives the data as company T.O data, and the crew checks and inserts it, as in FCOM DSC-22-FMS-20-30.

The flyPad screen

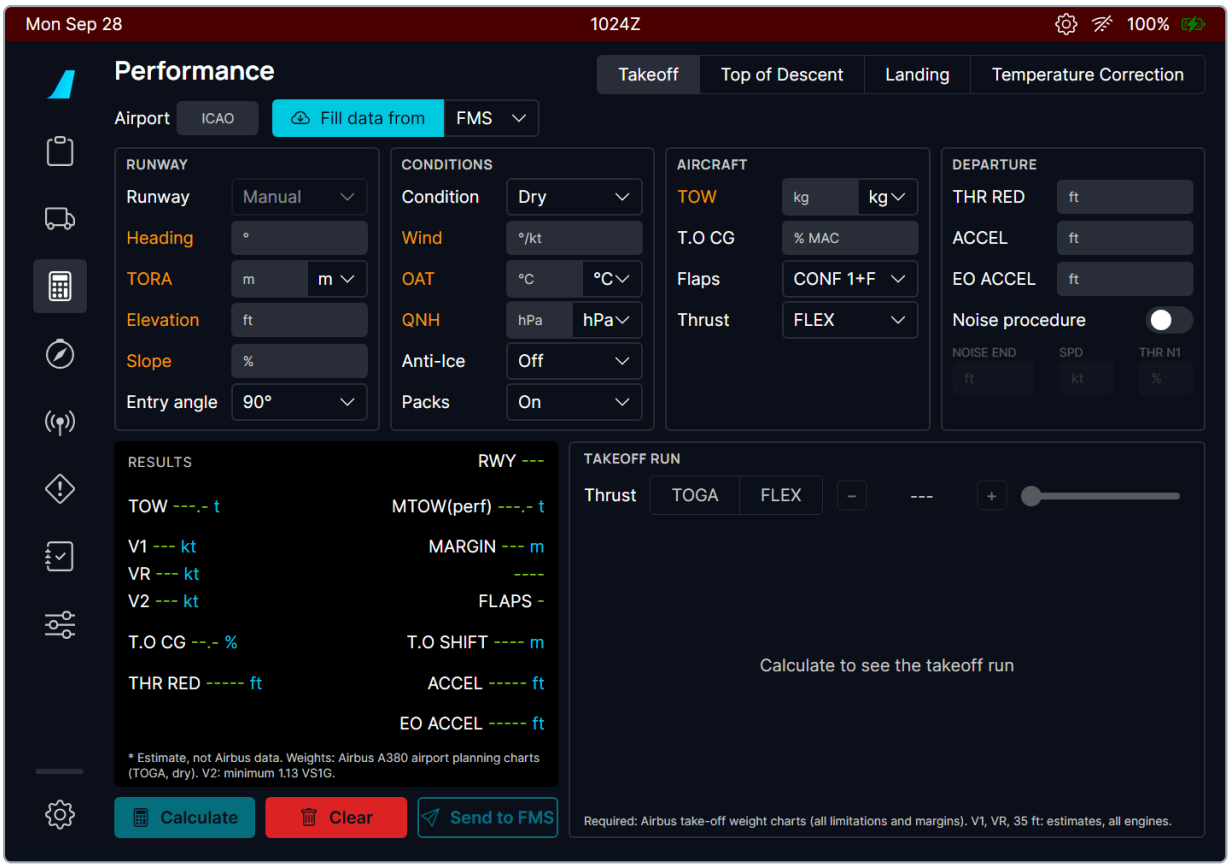


Figure 1. The takeoff screen. Amber labels: inputs still missing (Calculate stays dim).

Part	Contents
Airport, Fill data from	ICAO code; imports from the **FMS** (departure, runway, TOW, CG, THR RED / ACCEL), the **OFP** (SimBrief) or the **METAR** (wind, OAT, QNH).
RUNWAY	Runway from the list (or Manual), heading, TORA, elevation, slope, and the **entry angle** (90° or 180° turn onto the runway, or 0° when rolling straight on: it sets the line-up distance lost).
CONDITIONS	Runway condition, wind, OAT, QNH, anti-ice, packs.
AIRCRAFT	TOW in t (or klb), one decimal, T.O CG in % MAC, flaps (CONF 1+F, 2, 3 or OPT = the best one), thrust (TOGA, FLEX or DERATED: see "A380 takeoff calculator: DERATED takeoff" below).
DEPARTURE	THR RED, ACCEL, EO ACCEL altitudes (they default to the elevation + 1500 ft), and the noise abatement procedure (NOISE END altitude, speed, N1).
RESULTS	The results in the layout of the OIS takeoff application (FCOM PER-TOF-TOR-SRS).
TAKEOFF RUN	The runway with the V1, VR and 35 ft points, the required distance and the margin, for TOGA, any FLEX temperature of the allowed range, or a DERATED level the TOW permits.

A FLEX takeoff

1. Fill the airport and runway (or **Fill data from** FMS / OFP), the conditions (METAR), the TOW and the T.O CG.
2. Keep **Thrust FLEX** and press **Calculate**.
3. Read the FLEX temperature and the speeds in RESULTS; check the takeoff run and its margin.
4. **Send to FMS**, then insert the data on the MFD (see "On the MFD: company T.O data" below).

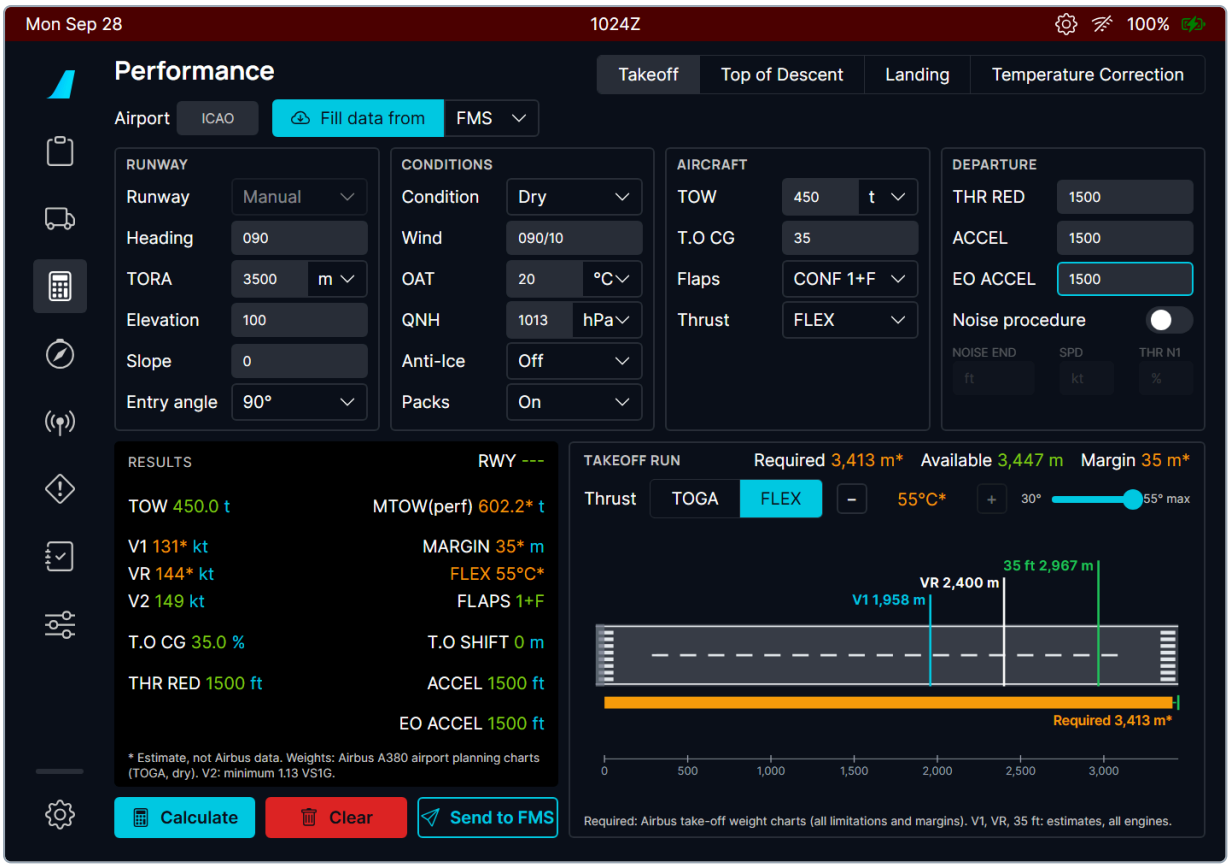


Figure 2. 450 t, 3500 m, 20 °C, CONF 1+F: FLEX 55 °C (the maximum, amber: estimate), V1 131 / VR 144 / V2 149 kt, MTOW(perf) 602 t. The takeoff run needs 3413 m of the 3447 m available after line-up: 35 m margin.

Reading the results

Line	Meaning
TOW / MTOW(perf)	Your takeoff weight, and the maximum the runway and conditions allow at TOGA.
V1, VR, V2	The takeoff speeds. V2 is at least 1.13 VS1G and 1.10 VMCA, VR at least 1.05 VMCA, V1 at least VMCG (FCOM PER-TOF-TOR-SRS).
MARGIN	Runway left beyond the required distance.
FLEX / TOGA	The FLEX temperature, or TOGA.
FLAPS, T.O CG, T.O SHIFT	The configuration, the CG entered, the takeoff shift (distance lost at line-up or an intersection).
THR RED, ACCEL, EO ACCEL	The altitudes of the DEPARTURE section.

> **NOTE**
> **Green** values come from Airbus data (the takeoff weight charts of the A380 Aircraft Characteristics); **amber with a star** are estimates of the FBW model (see Limitations).

TOGA or another FLEX temperature

The TAKEOFF RUN panel lets you look at the run for TOGA, or for any FLEX temperature between the OAT and the maximum (FCOM PER-TOF-THR-FLX). Use **-** / **+** or the slider; the speeds and the run follow.

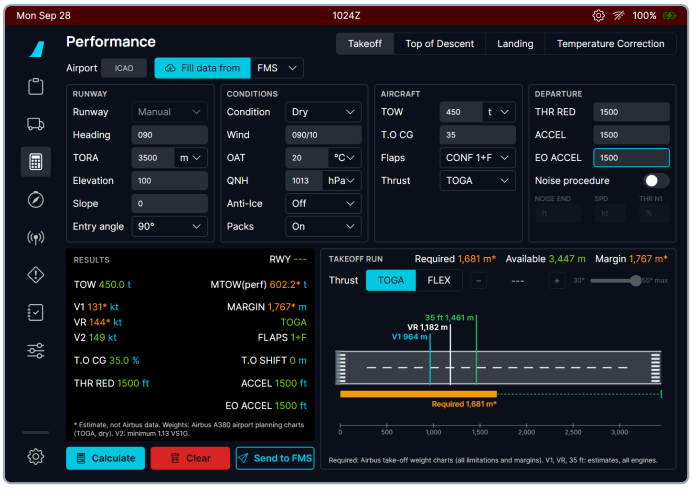


Figure 3. TOGA: the run needs 1681 m, margin 1767 m.

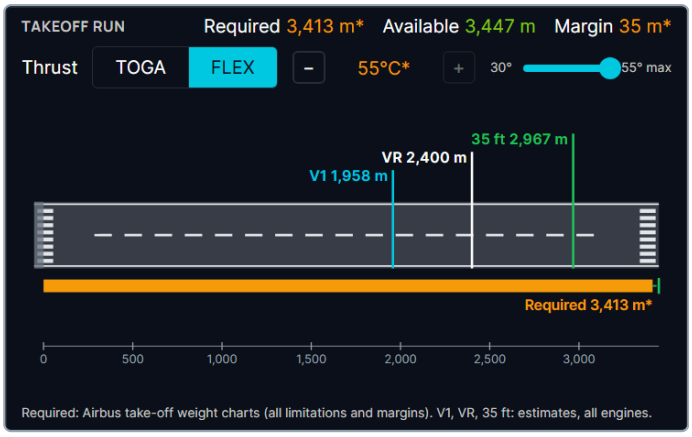


Figure 4. FLEX range 30 °C to 55 °C max: the selected temperature is the one sent to the FMS.

Send to FMS sends the thrust selected for the takeoff run only (TOGA, FLEX at its temperature or DERATED at its level), with its own speeds.

Options

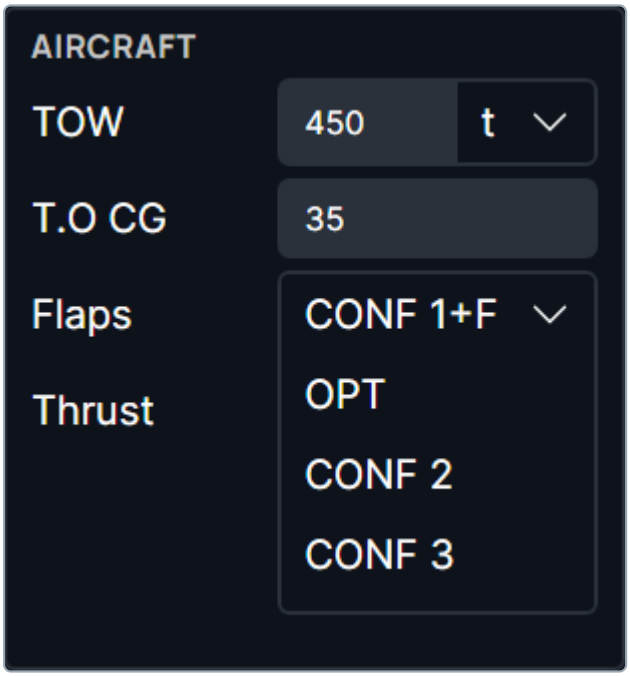


Figure 5. Flaps: CONF 1+F, 2, 3, or OPT (the calculator picks the best).

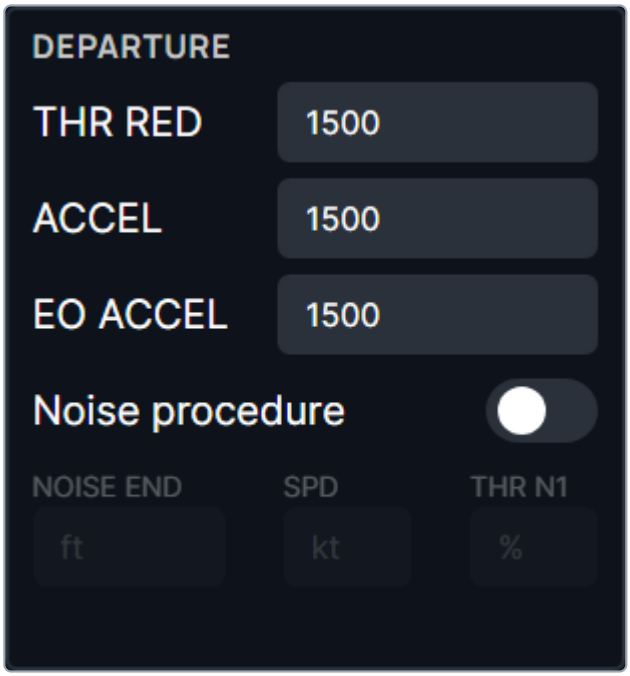


Figure 6. Noise procedure: NOISE END altitude, speed and N1 become available; they are sent with the data.

Errors

When the takeoff is not possible, a message says why and there is no result:



Figure 7. CG outside the takeoff envelope (here 30 % at 450 t: the forward limit moves aft with the weight).



Figure 8. 510 t on 2100 m at 40 °C: the maximum takeoff weight is 438.8 t.



Figure 9. A wet or contaminated runway: the A380 data covers dry runways only.

Message	Meaning
Beyond Takeoff Weight Limit - Max ...	TOW above the runway limit: reduce the weight or use another runway.
CG Outside Takeoff Envelope	The CG is outside the A380 takeoff CG envelope for this weight.
Above Structural MTOW	TOW above 510 t.
Only dry runways are covered...	The A380 data has no wet or contaminated takeoff.
Runway shorter than the A380 performance data (1700 m)	The charts start at 1700 m.
Above Maximum Tailwind / OAT / Pressure Altitude	Outside the A380 limits (FCOM LIM-12).
Flex Not Possible	No FLEX temperature is possible at this weight: take off at TOGA.

On the MFD: company T.O data

The MFD PERF T.O page gets two buttons: **CPNY T.O REQUEST** (ask the airline) and, once data has arrived, **RECEIVED CPNY T.O**.

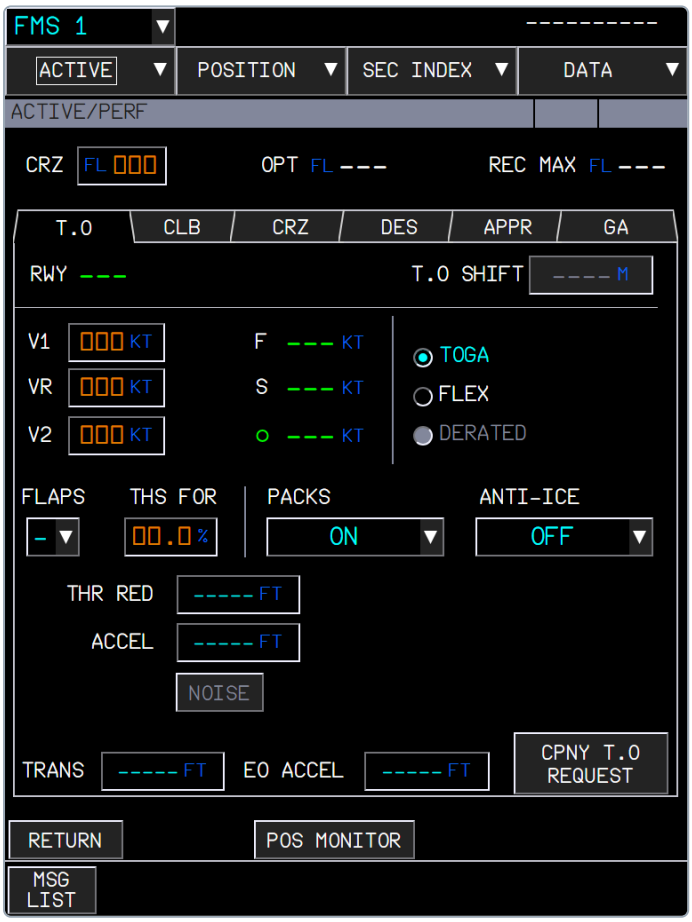


Figure 10. PERF T.O before any data: the CPNY T.O REQUEST button (bottom right).



Figure 11. COMPANY T.O DATA REQUEST: up to 2 runways, TOGA / FLEX / DERATED, free text; SEND T.O REQUEST.

- After SEND T.O REQUEST the page shows *REQUEST PENDING...*; without an answer after 4 minutes, *NO COMPANY REPLY*. The flyPad **Send to FMS** is that answer.
- When data arrives: the message *COMPANY T.O DATA RECEIVED WAITING FOR INSERTION* and the button **RECEIVED CPNY T.O** on PERF.

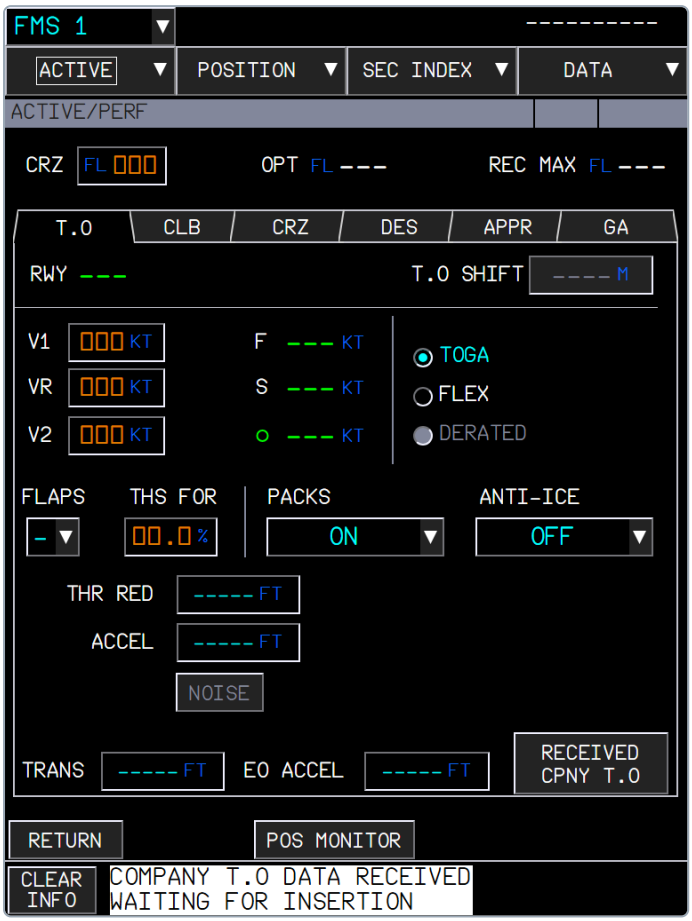


Figure 12. Data received: the message and the RECEIVED CPNY T.O button.

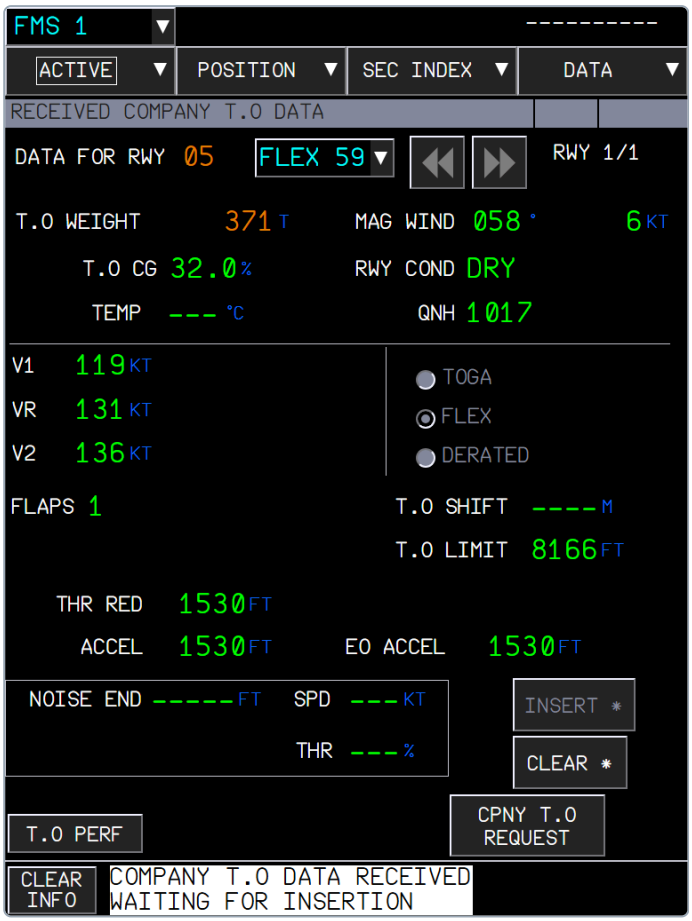


Figure 13. RECEIVED COMPANY T.O DATA: runway 05 in amber (not the flight plan runway), TOW in amber with UPLINK/ACTIVE TOW DISAGREE: INSERT is not available.

The RECEIVED COMPANY T.O DATA page

- Up to 4 runways (arrows, RWY 1/4) × 2 thrust ratings (the TOGA / FLEX list at the top).
- **Runway in amber**: not the departure runway of the active flight plan.
- **TOW in amber** with *UPLINK/ACTIVE TOW DISAGREE*: the uplink TOW is more than 2 t below or 7 t above the FMS TOW (FUEL&LOAD).
- TEMP shows dashes for a FLEX or DERATED takeoff (a derate reads `DRT D12` in the thrust list); T.O LIMIT is the TORA; the noise fields are filled when the procedure was sent.
- **INSERT** (only when runway and TOW agree) fills PERF T.O: speeds, flaps, thrust, shift, THR RED, ACCEL, EO ACCEL and noise. It does not fill THS FOR: the crew enters the CG there. **CLEAR** deletes the data.

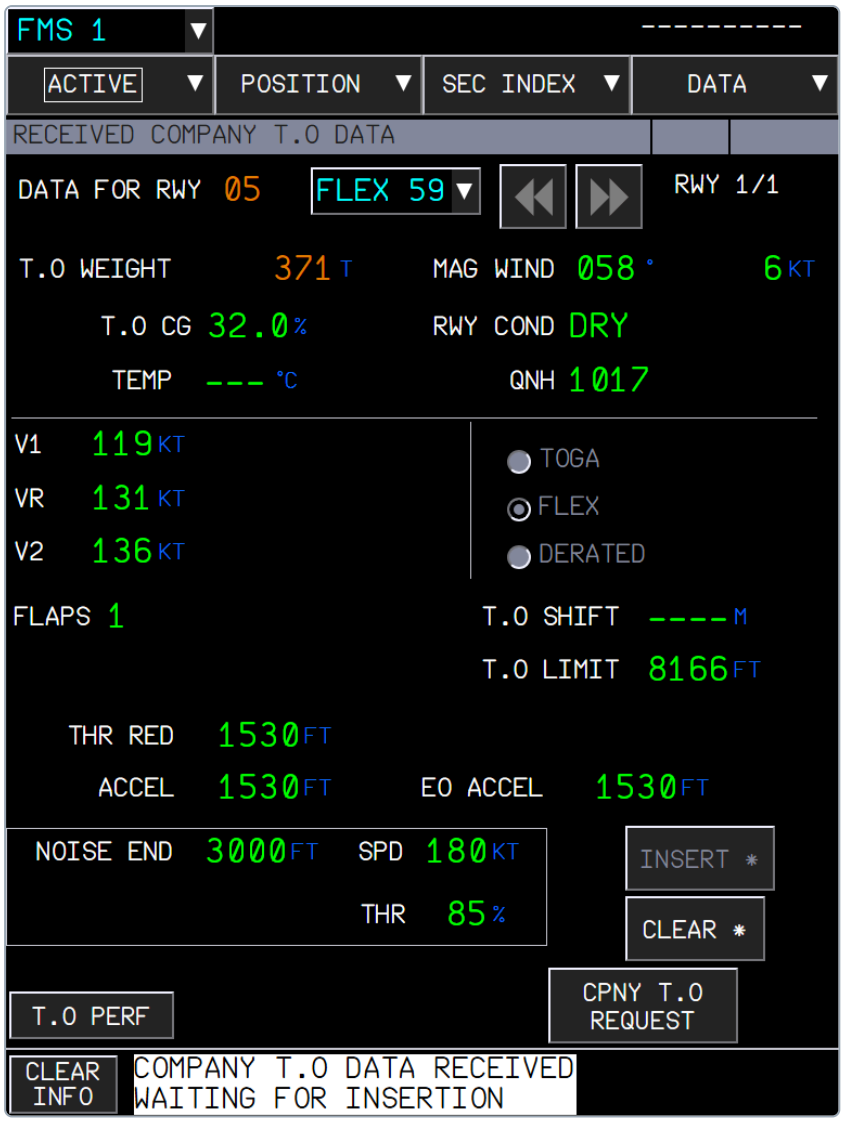


Figure 14. An uplink with the noise procedure: NOISE END 3000 ft, SPD 180 kt, THR 85 %.

> **TIP**
> The flyPad warns you right after Send to FMS if INSERT will not work: another departure runway in the FMS, no TOW in the FMS, or a TOW outside -2 t / +7 t of the FMS one.

What to expect

Situation	What you should see
Inputs missing	Amber labels; Calculate stays dim.
Calculate with valid inputs	RESULTS and the takeoff run; FLEX at its maximum by default.
TOW too high, CG out of envelope, wet runway...	A message with the reason; no result.
Send to FMS with a runway from the list	"Takeoff data sent to the FMS: RECEIVED CPNY T.O on the PERF page"; the MFD shows the data a moment later.
Send to FMS with a manual runway	Nothing is sent: the FMS needs the runway ident.
RECEIVED page, other runway than the flight plan	Runway in amber; INSERT not available.
INSERT	PERF T.O filled; THS FOR still to enter.

Limitations and why

Limitation	Why	What to do
Dry runways only (the DERATED takeoff too: see its own limitations below).	The only public A380 takeoff data is the takeoff weight charts of the Airbus A380 Aircraft Characteristics (dry, ISA and ISA+15, runway 1700-5100 m, altitude 0-8000 ft).	On a wet runway, use the dry result with extra margin, as a training estimate only.
Speeds, FLEX, wind, slope and temperature effects are estimates (amber *).	The charts give weights only. The rest is a model fitted on them and checked on the FCOM examples (FLEX 38 at 560 t / 614 t).	-
The run distances (V1, VR, 35 ft points) are all-engine estimates.	No A380 distance data is public.	-
Send to FMS needs a runway from the list.	The FMS checks the uplink against its departure runway.	Choose the runway in the list.
Not certified data.	For simulation and training.	-

How to use

1. On the MFD, enter the flight plan with its departure runway and fill FUEL&LOAD (ZFW, ZFWCG, block).
2. flyPad > **Performance** > **Takeoff**: **Fill data from** FMS, then METAR (or type the data).
3. Check TOW, T.O CG, flaps and thrust; press **Calculate**.
4. In TAKEOFF RUN, choose TOGA, the FLEX temperature or the DERATED level you will use; press **Send to FMS**.
5. On the MFD PERF T.O page: **RECEIVED CPNY T.O**, check the page, **INSERT**; enter the T.O CG in THS FOR.

How to test

#	Do this	Expected result
1	A380X on the ground, ADIRS aligned. MFD: flight plan with a departure runway, FUEL&LOAD filled. Open flyPad > Performance > Takeoff.	Empty screen: amber labels on the missing inputs, Calculate dim.
2	Choose FMS in **Fill data from** and press it, then METAR.	Airport, runway, TOW (in t, one decimal), CG, THR RED / ACCEL filled; wind, OAT and QNH from the METAR.
3	Press **Calculate**.	RESULTS: TOW, MTOW(perf), V1 / VR / V2, FLEX (or TOGA), flaps; chart values green, estimates amber with a star; TAKEOFF RUN with the V1 / VR / 35 ft marks and the margin.
4	On a long runway (e.g. 4000 m near sea level, 15 °C), type TOW 510.0 t with T.O CG 35.75 %, then 43 %, and Calculate each time.	Both accepted (envelope edges included).
5	T.O CG 35.7 % at 510 t; then 43.5 % at 400 t.	Message "CG Outside Takeoff Envelope", no result, each time.
6	Set the runway condition to Wet and Calculate.	Message that only dry runways are covered; no result.
7	Back to dry. In TAKEOFF RUN press TOGA, then FLEX and move the slider.	The run, the speeds and the margin follow; the FLEX range runs from the OAT up to the maximum.

8	Press **Send to FMS**.	Toast "Takeoff data sent to the FMS: RECEIVED CPNY T.O on the PERF page". After the transit time of the Company Datalink Reply Time setting (Real: 5 to 15 s), the MFD shows COMPANY T.O DATA RECEIVED WAITING FOR INSERTION and the RECEIVED CPNY T.O button on PERF T.O.
9	MFD PERF T.O > **RECEIVED CPNY T.O** > **INSERT**.	PERF T.O gets the speeds, flaps, thrust, T.O shift, THR RED / ACCEL / EO ACCEL (and the noise data when sent); THS FOR stays empty.
10	Change the FMS departure runway, then Send to FMS again.	flyPad warning about the runway; on the RECEIVED page the runway is amber and INSERT is not available.
11	PERF T.O > **CPNY T.O REQUEST** > **SEND T.O REQUEST**, and send nothing from the flyPad.	REQUEST PENDING..., then NO COMPANY REPLY after 4 minutes.

2. A380 takeoff calculator: DERATED takeoff

> **Aircraft:** A380X (the A32NX has no derate) | **Where:** flyPad > Performance > Takeoff > TAKEOFF RUN; MFD PERF T.O | **Status:** Tested in the sim by the user: OK (flyPad to MFD as DRT D08 and INSERT; E/WD D12 with N1 at the derated limit, 83.0 % against 88.3 % at TOGA; PFD FMA MAN DTO and the ENG THR LEVERS NOT SET caution at TOGA). Checked in the browser harness. Unit tests.

What it is for

A derate is a lower takeoff rating that is certified on its own (A380 FCOM PER-TOF-THR-DRT): the flight manual gives it its own takeoff limits and performance. The A380 has six levels, **D04, D08, D12, D16, D20, D24**: each one takes about 4 % more off the maximum takeoff thrust (D04 = about 4 % less, D24 = about 24 % less). The MFD PERF T.O page and the engines already had the derate; the flyPad calculator offered only TOGA and FLEX.

A derate is not FLEX. FLEX uses an assumed temperature and the crew can still push the levers to TOGA. A derated thrust depends on the real outside temperature, and the crew may not set TOGA during a derated takeoff below the speeds of the engine failure after V1 technique (the only exception is the windshear recovery). So the FCOM expects FLEX to give the better margin in most cases, and keeps the derate for **limiting runways**. The lower thrust also lowers the minimum control speeds VMCG and VMCA.

What you see

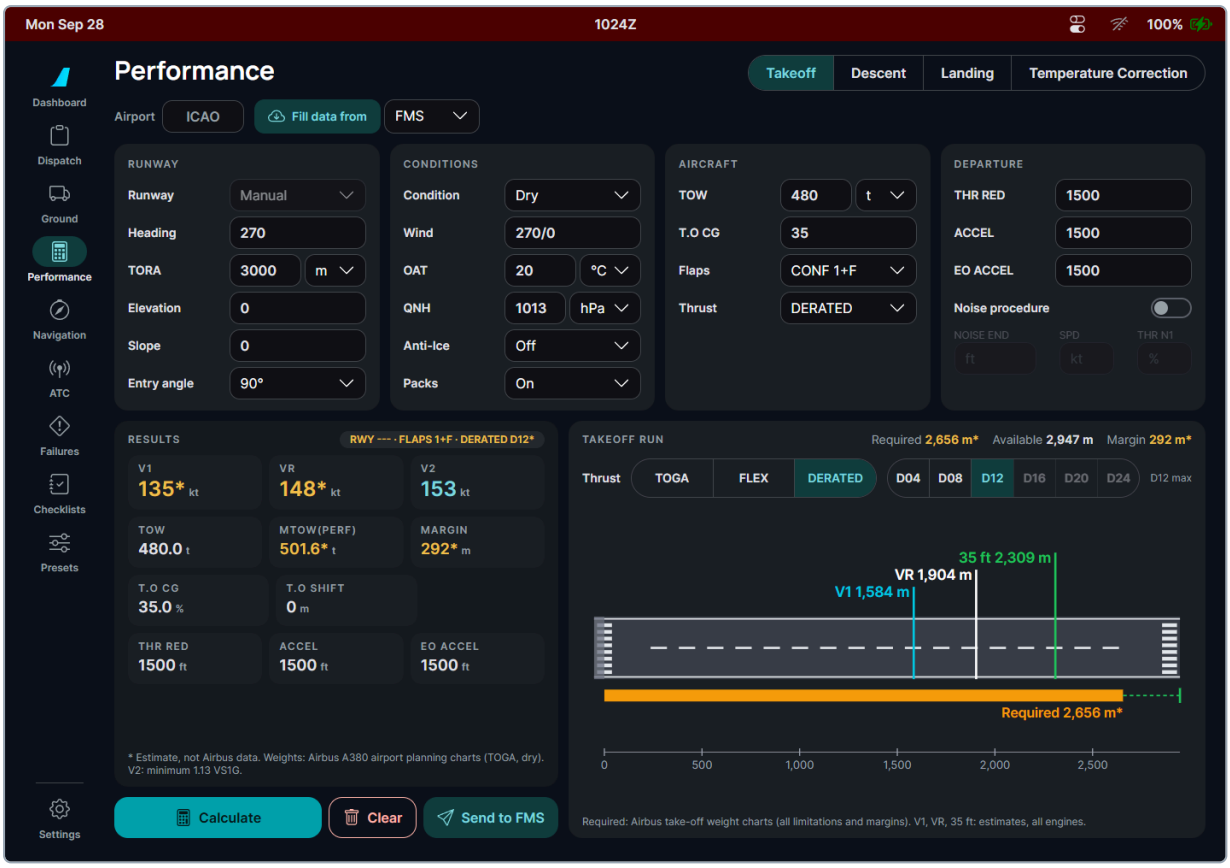


Figure 15. 480 t, 3000 m, sea level, 20 °C, CONF 1+F. TAKEOFF RUN > **DERATED**: D12 is chosen, the deepest level this TOW permits (D16, D20, D24 greyed). RESULTS reads DERATED D12*, MTOW(perf) 501.6* t is the D12 limit (the same take-off at FLEX: 44 °C, MTOW(perf) 570 t).

- **Three exclusive choices.** TOGA, FLEX and DERATED are three buttons of the TAKEOFF RUN panel: a derate and FLEX cannot be combined (FCOM: on the aircraft too, the MFD takes either a level or a FLEX temperature). The *Thrust* input of the AIRCRAFT section follows the same choice, both ways.
- **The level picker** D04 to D24 replaces the FLEX slider. The levels the TOW does not permit are greyed. DERATED starts at the **deepest permitted level**; the deepest one is written at the right ("D12 max").
- **RESULTS** shows `DERATED D12*` in its header; with a shallower level than the deepest one it adds the maximum, e.g. `DERATED D04* (max D12)`. The speeds, MTOW(perf), the margin and the run are those of the chosen level.
- If DERATED is chosen and no level is possible at this weight, the run goes back to TOGA with the message **Derate Not Possible**.

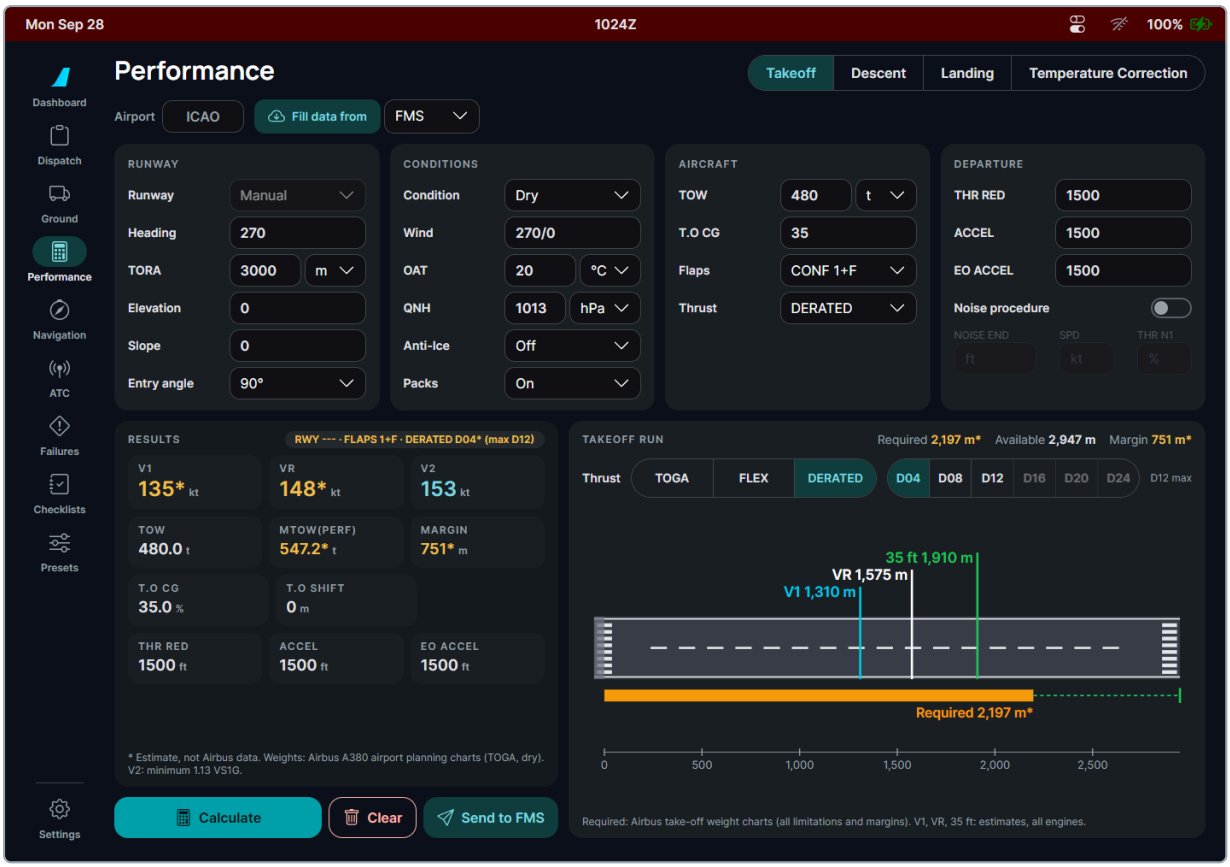


Figure 16. D04 picked by hand: header DERATED D04* (max D12), MTOW(perf) 547.2* t, the run needs 2197 m of the 2947 m available.

How it is calculated

All derate values are estimates of the FBW model (amber, with a star):

- **Weight limit** = the TOGA weight limit at the actual OAT x (1 - level %). In the pictures: TOGA limit 570 t, so D04 = 570 x 0.96 = 547.2 t and D12 = 570 x 0.88 = 501.6 t. In DERATED mode MTOW(perf) shows this derated limit.
- **Default level** = the deepest level whose limit is still at or above the TOW (D12 for 480 t above: D16 would allow only 478.8 t).
- **Speeds**: as for TOGA, with VMCG and VMCA lowered by the square root of the thrust ratio. The FCOM says that a lower thrust lowers them but gives no values: this factor is our estimate. When the speeds are set by the stall margin (V2 at least 1.13 VS1G), they do not change.
- **Takeoff run**: the V1, VR and 35 ft distances are computed for the thrust of the level.

Runway condition

FLEX is not permitted on a contaminated runway (FCOM PER-TOF-CTA); a derate is permitted on dry, wet and contaminated runways. So on a contaminated runway the flyPad offers **TOGA and DERATED** only. The A380 calculator still computes **dry runways only**: wet and contaminated runways still show the "runway condition not supported" message (see Limitations).

On the MFD: DERATED company T.O data

- **Send to FMS** with DERATED sends the thrust DERATED and its level with the speeds of that level.
- The RECEIVED COMPANY T.O DATA page shows the thrust as `DRT D12` (TEMP: dashes).
- **INSERT** sets PERF T.O to DERATED with the level. The engines then use the derated limit: the E/WD shows the level (e.g. D12) and the takeoff N1 is lower than the TOGA one.
- A **CPNY T.O REQUEST** for DERATED carries no level (the request page has none): the flyPad then picks the deepest permitted level.

- A320: no derate. The A320 calculator never offers it, and the MCDU refuses a DERATED uplink with *INVALID TAKEOFF UPLINK*.

On the PFD and the ECAM during the takeoff

- **FMA**: with a derated takeoff and the thrust levers at FLX/MCT, the first column of the PFD FMA shows `MAN` over `DTO` in a white box (not `MAN FLX +XX`, which is for a FLEX takeoff). This needs the fix e54ede63e (FMA shows MAN DTO for a derated takeoff).
- **No TOGA**: TOGA is not allowed for a derated takeoff (FCOM PRO-ABN-ECAM-10-70). With a derate selected, thrust levers at TOGA, even for a moment (e.g. pushed through TOGA on the way to FLX/MCT), trigger the master caution **ENG THR LEVERS NOT SET**, as on the real A380. The ECAM action is `THR LEVERS ... MCT/FLEX`. So set the levers straight to FLX/MCT.

Limitations and why

Limitation	Why	What to do
The derate is computed for dry runways only.	The Airbus charts behind the calculator are dry only, and the FCOM has no wet or contaminated A380 takeoff numbers (only the rules and the crosswind table).	Wet and contaminated runways show "runway condition not supported": no result.
Derate speeds, weight limits and distances are estimates (amber *).	The FCOM gives the levels and the rules, not the performance per level; the VMC decrease factor is our own.	Use them for simulation and training only.
No "optimum derate" for a VMCG-limited takeoff.	The FCOM caution not to use a derate deeper than the one giving the highest MTOW(perf) applies to VMCG-limited takeoffs. The chart model has no VMCG-limited weight, so it cannot find that optimum: the deepest level is simply the deepest one the TOW permits.	-

How to use

1. Calculate the takeoff as usual (see the A380 takeoff calculator above).
2. In TAKEOFF RUN press **DERATED** (or choose DERATED in the AIRCRAFT *Thrust* input): the deepest permitted level is selected.
3. Pick another level if you want a smaller reduction; check MTOW(perf), the speeds and the margin.
4. Press **Send to FMS**, then on the MFD PERF T.O: **RECEIVED CPNY T.O**, check `DRT Dnn`, **INSERT**.
5. For the takeoff, set the thrust levers **straight to FLX/MCT**: TOGA is not allowed with a derate, and even passing through TOGA triggers ENG THR LEVERS NOT SET. The PFD FMA shows MAN DTO.

How to test

#	Do this	Expected result
1	A380X on the ground. flyPad > Performance > Takeoff: heading 270, TORA 3000 m, elevation 0, slope 0, wind 270/0, OAT 20 °C, QNH 1013, dry, TOW 480 t, T.O CG 35 %, **Calculate**.	FLEX 44 °C, MTOW(perf) 570 t.
2	In TAKEOFF RUN press **DERATED**.	Level picker D04 to D24 with D12 selected and D16, D20, D24 greyed; "D12 max"; RESULTS header DERATED D12*; MTOW(perf) 501.6* t; AIRCRAFT Thrust reads DERATED.

3	Press **D04**.	Header DERATED D04* (max D12); MTOW(perf) 547.2* t; a shorter run.
4	Press FLEX, then DERATED again.	FLEX shows the slider (44 °C max); DERATED comes back at D12.
5	Set the runway condition to a contaminated one and open the AIRCRAFT Thrust list.	TOGA and DERATED only (no FLEX); Calculate still gives the "runway condition not supported" message.
6	Back to dry, DERATED D08, with a runway from the list and the flight plan on the MFD: **Send to FMS**.	After the reply time: MFD RECEIVED COMPANY T.O DATA with the thrust `DRT D08`, TEMP dashes.
7	**INSERT**, then look at PERF T.O and, with the engines running and the levers at FLX/MCT, the E/WD.	PERF T.O DERATED D08; the E/WD shows D08 and the takeoff N1 limit is lower than at TOGA.
8	Ready for takeoff with the derate inserted: set the thrust levers **straight to FLX/MCT** (not through TOGA) and look at the PFD FMA.	First FMA column: `MAN` / `DTO` in a white box (not MAN FLX +XX). No caution.
9	Still with the derate selected, push the thrust levers to **TOGA** (even briefly).	Master caution and the E/WD alert **ENG THR LEVERS NOT SET** with `THR LEVERS ... MCT/FLEX`.
10	A32NX: flyPad > Performance > Takeoff.	Only TOGA and FLEX; no DERATED.

3. A320 takeoff calculator and MCDU uplink takeoff data

> **Aircraft:** A32NX | **Where:** flyPad > Performance > Takeoff; MCDU PERF TAKE OFF > <TO DATA (6L) | **Status:** The flyPad calculation was used in the sim; the MCDU uplink pages are checked in the browser harness only; the full flow is not sim-tested. Unit tests of the takeoff distances.

What it is for

The A320 takeoff speeds, FLEX temperature and THS come from the airline or the crew's own tool. Here the flyPad computes them with the FlyByWire A320neo performance model (built from Airbus data) and sends them to the FMGC as an **uplink takeoff data** message, which the crew inserts from the MCDU PERF TAKE OFF page as on the real aircraft.

The flyPad screen (A320)

The screen is the one of the A380 (see "A380 takeoff calculator and company T.O data" above) with these A320 differences:

- No noise procedure fields; T.O CG is optional: below 27 % MAC the forward CG performance applies (A320 FCOM PER-LOD) and the THS is shown in RESULTS.
- All values come from the FBW A320 model: they are shown in green (no estimates).
- Wet and contaminated runways are computed (the A380 page accepts dry only).

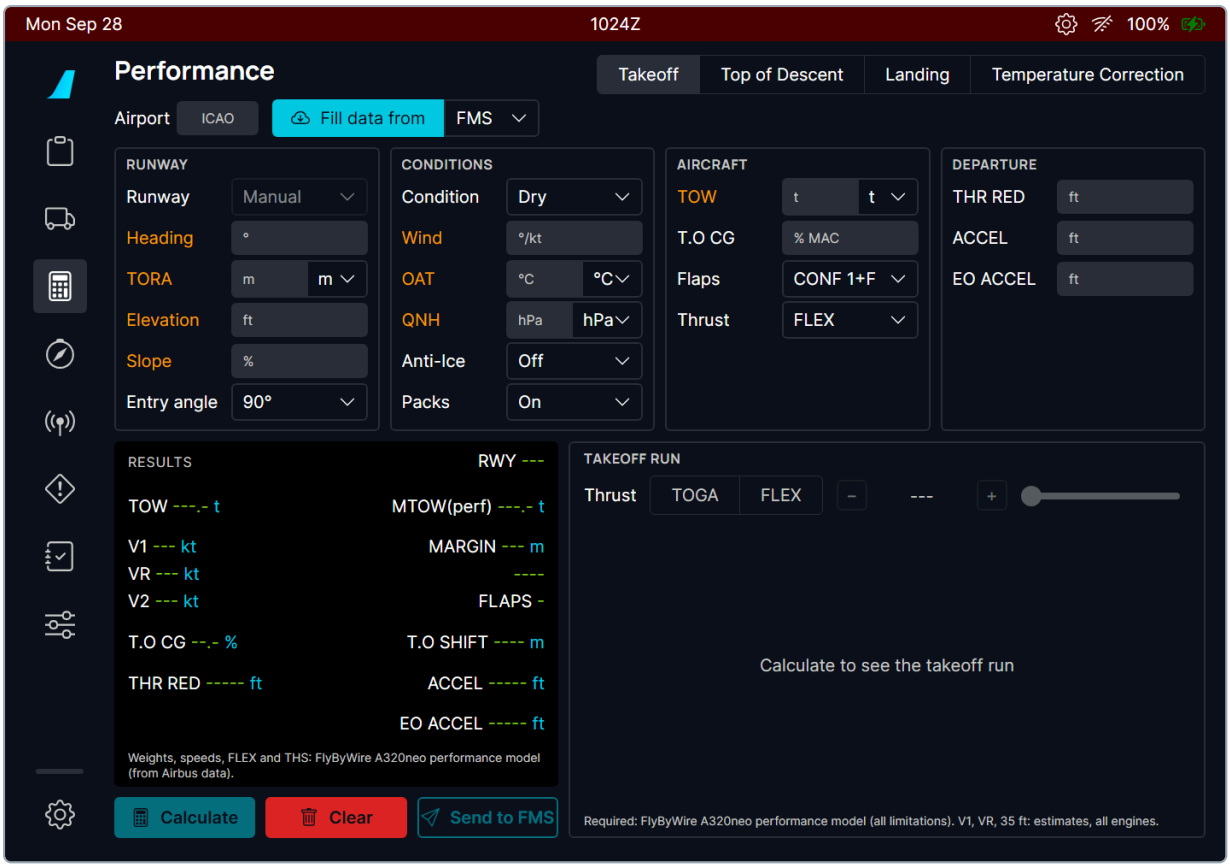


Figure 17. The A320 takeoff screen before any input.

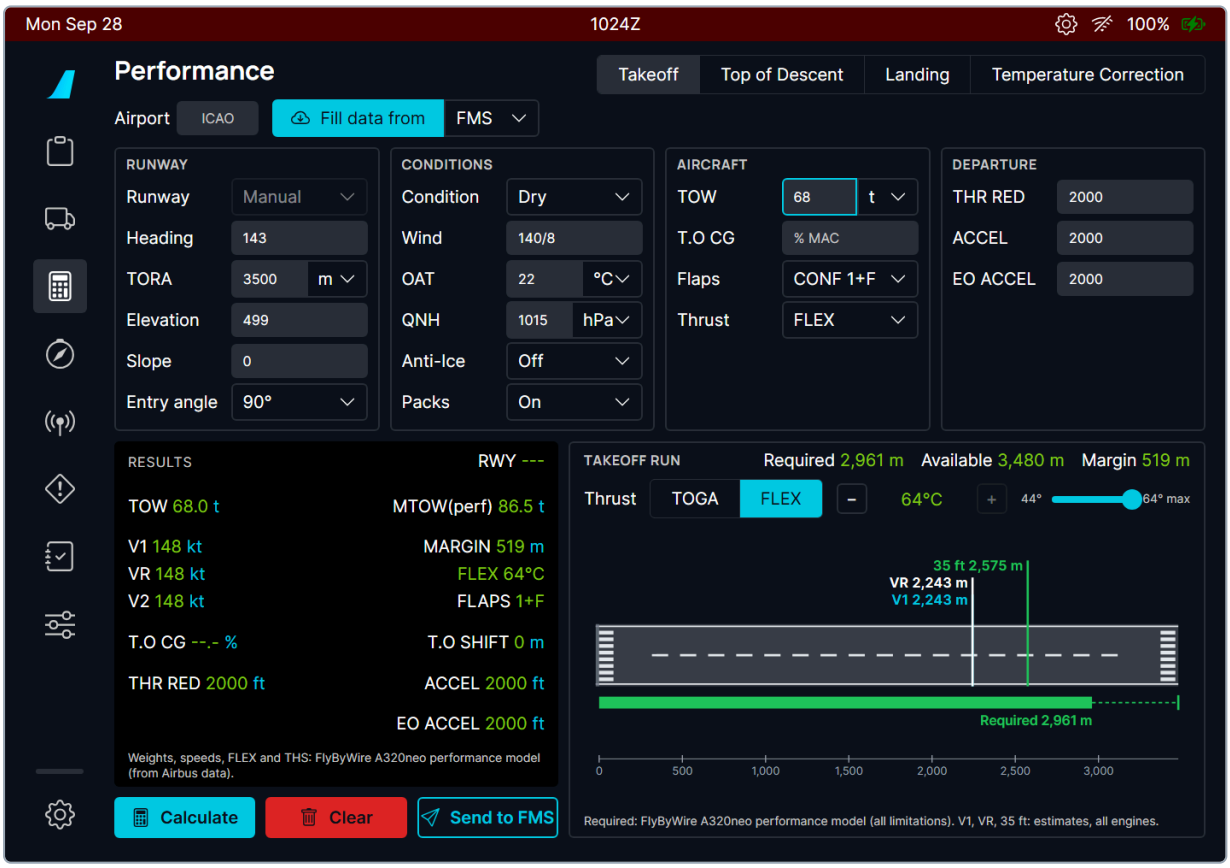


Figure 18. LFBO 14R-like runway (3500 m, 499 ft), 68 t, 22 °C: FLEX 64 °C (maximum), required 2961 m of 3480 m, margin 519 m.

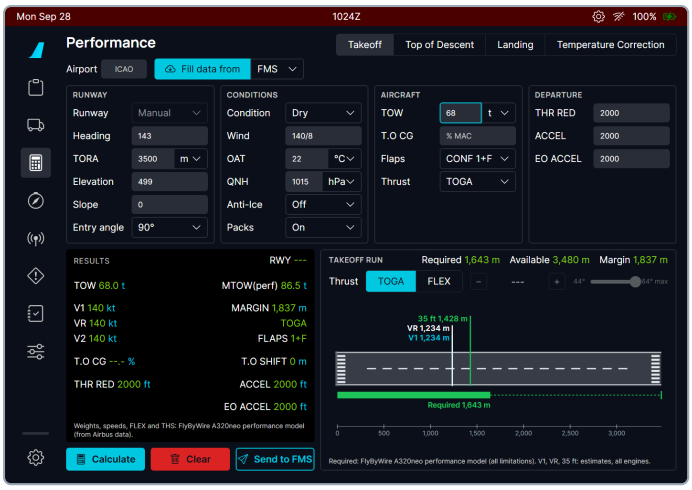


Figure 19. The TOGA run at the same weight: 1643 m.

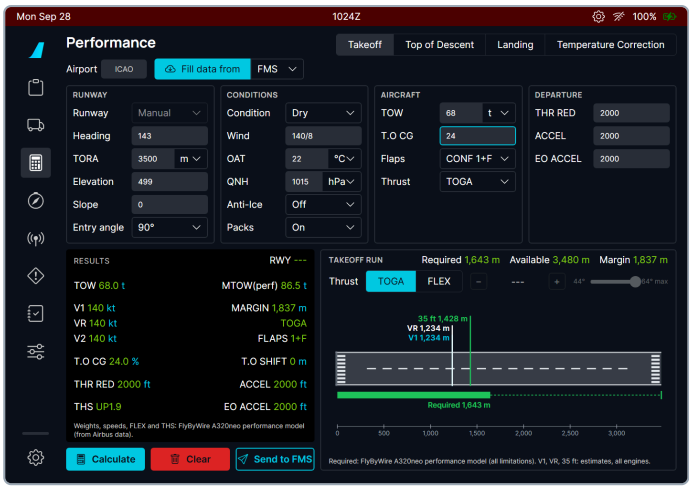


Figure 20. T.O CG 24 % (forward CG) at TOGA: the THS (UP1.9) appears in RESULTS.

> **NOTE**
>
> At light weights the FBW A320 model gives **V1 = VR = V2** (the pictures show 148/148/148 at FLEX and 140/140/140 at TOGA for 68 t). This comes from the FlyByWire model itself (the same with the FBW version before this change), not from this feature.

Sending to the MCDU

1. On the flyPad, choose the **runway from the list** (the uplink carries the runway ident), calculate, choose TOGA or the FLEX temperature for the takeoff run, and press **Send to FMS**.
2. The flyPad says "*Takeoff data sent to the FMS: MCDU PERF TAKE OFF, UPLINK TO DATA, RECEIVED TO DATA*", and warns right away if the FMGC departure runway or TOW will prevent the insertion.
3. On the MCDU the scratchpad shows **TAKEOFF DATA UPLINK**.
4. PERF TAKE OFF > **<TO DATA** (6L) > **RECEIVED <TO DATA** (6L): the UPLINK MAX TO DATA page (or FLX TO DATA with 4R). Check it, then **INSERT UPLINK**.

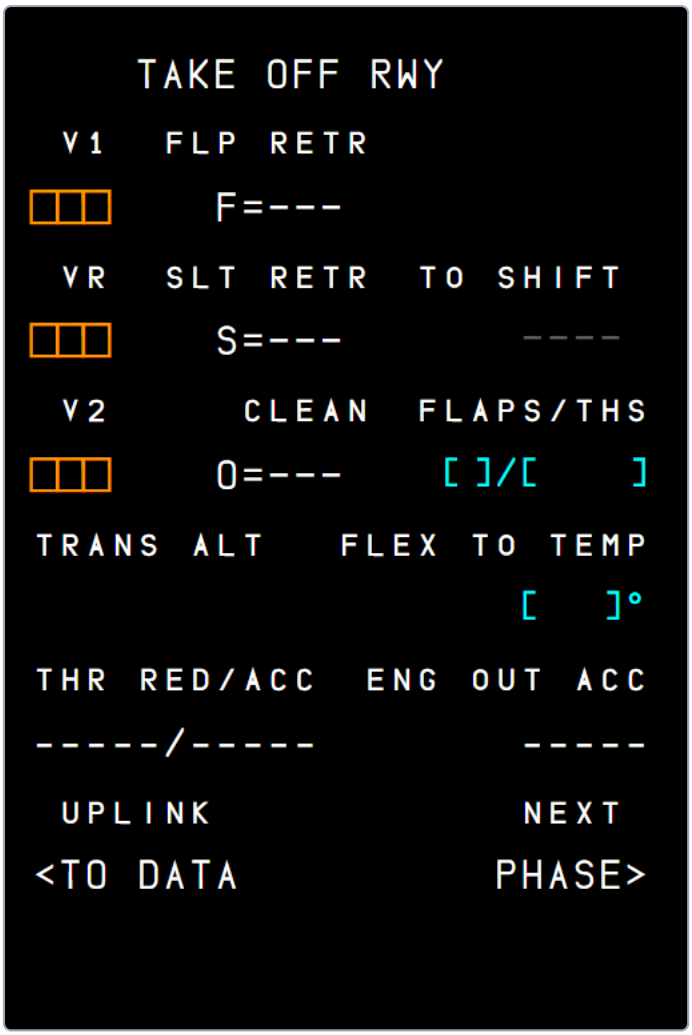


Figure 21. PERF TAKE OFF: the new UPLINK <TO DATA prompt (6L), in PREFLIGHT and DONE.

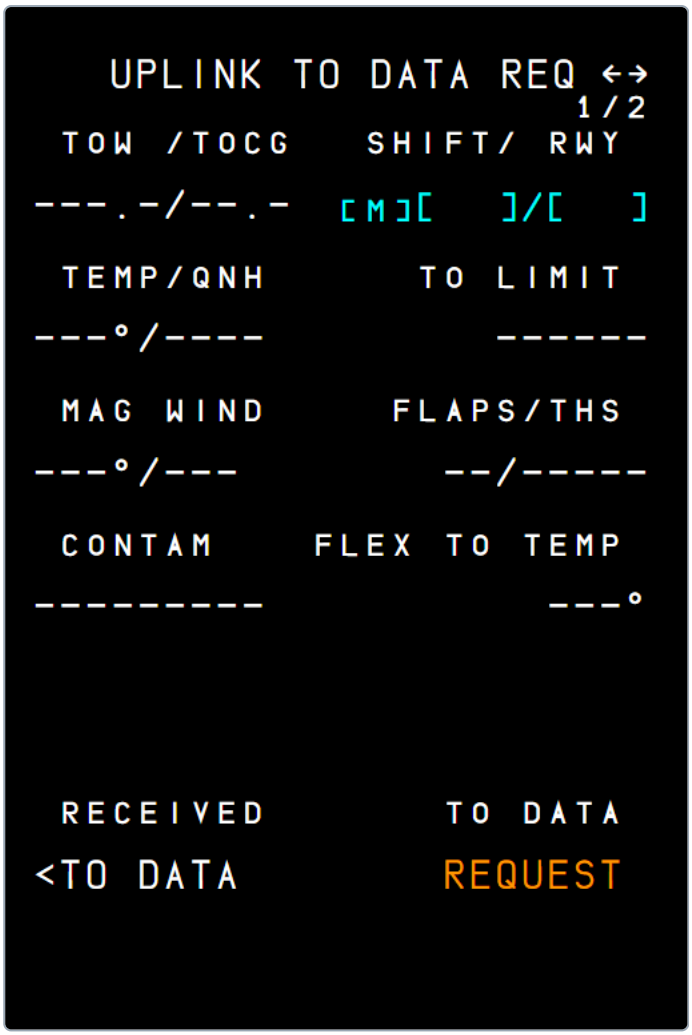


Figure 22. UPLINK TO DATA REQ: the request page (two runways, 1/2). RECEIVED <TO DATA (6L), TO DATA REQUEST (6R).

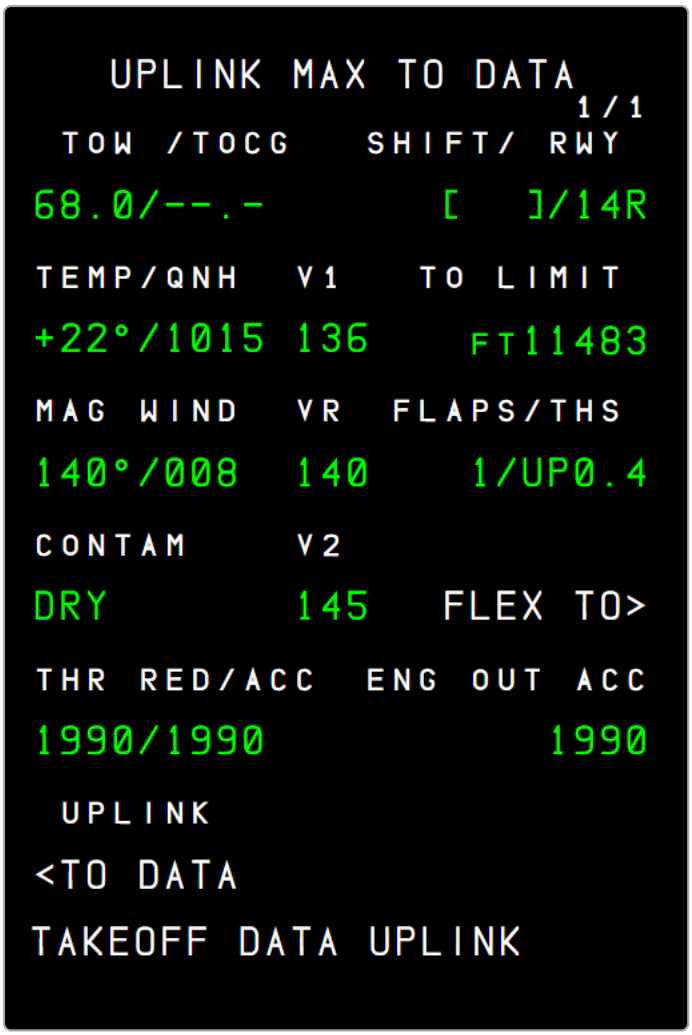


Figure 23. UPLINK MAX TO DATA after TAKEOFF DATA UPLINK: TOW, runway 14R, temperature and QNH, V1 / VR / V2, TO LIMIT, flaps / THS, THR RED / ACC, ENG OUT ACC. FLEX TO> (4R) shows the FLEX data.

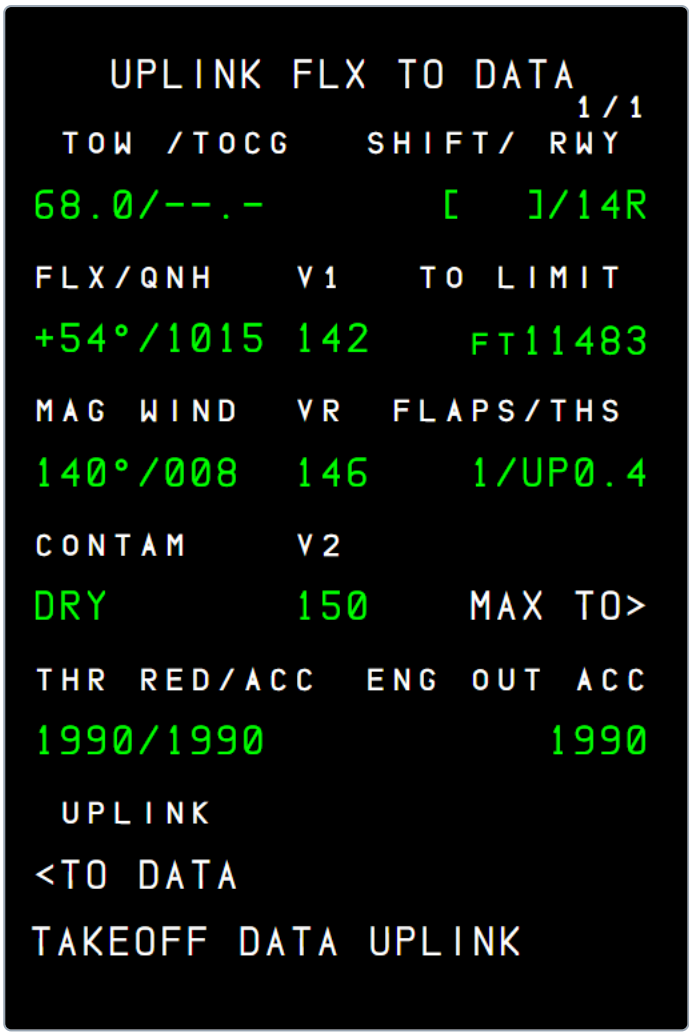


Figure 24. UPLINK FLX TO DATA: FLX +54 °C with its own speeds.

There is **no INSERT UPLINK prompt** in these pictures because the harness FMGC has no flight plan. The prompt appears only when the uplink is complete (V1 and VR present) and the checks below pass.

The insertion rules (A320 FCOM)

- The uplink runway must be the departure runway of the flight plan.
- The uplink TOW must be **not more than 1 t below or 3 t above** the FMGC TOW; otherwise there is no insertion.
- INSERT UPLINK fills V1, VR, V2, FLAPS/THS, FLEX TO TEMP (cleared for a MAX takeoff), THR RED/ACC, ENG OUT ACC and the TO SHIFT.
- An incomplete uplink (for example without V1 or VR) gives no INSERT UPLINK prompt (DSC-22_20-70 P 6).

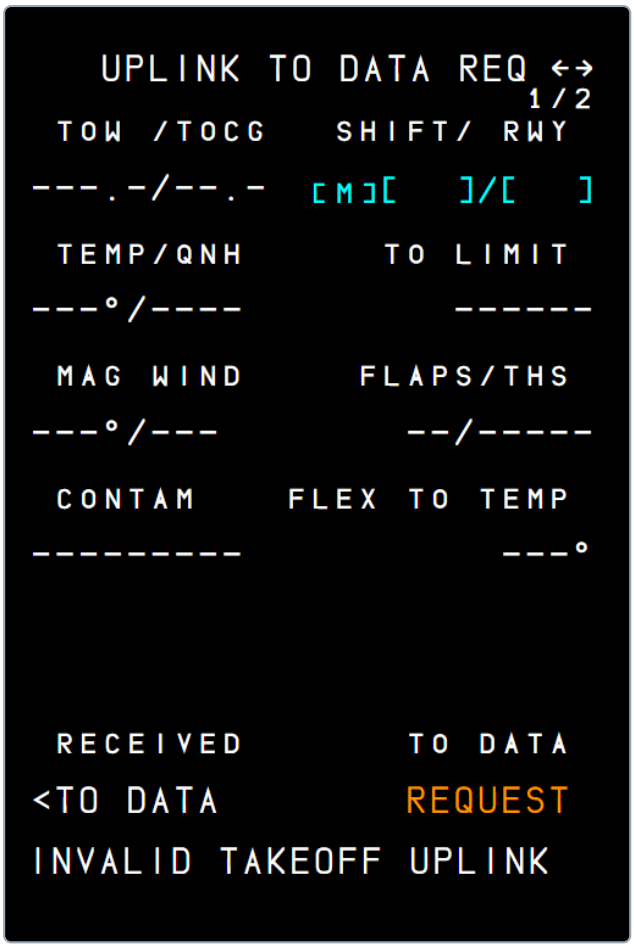


Figure 25. An uplink the FMGC cannot read: INVALID TAKEOFF UPLINK in the scratchpad.

MCDU message	Meaning
TAKEOFF DATA UPLINK	Uplink takeoff data received.
INVALID TAKEOFF UPLINK	The uplink is not valid (missing data).
REQUEST IS PENDING	A request was sent and has no answer yet.
NO ANSWER TO REQUEST	No answer 4 minutes after the request.

PERF TAKE OFF also gets the **TO SHIFT** field (1 m up to the runway length, CLR to delete): the distance lost when taking off from an intersection.

What to expect

Situation	What you should see
Send to FMS with a runway from the list	Success message; TAKEOFF DATA UPLINK on the MCDU.
Send to FMS with a manual runway	Nothing is sent: choose the runway in the list.
FMGC departure runway differs	flyPad warning; no INSERT UPLINK on the MCDU.
FMGC TOW differs by more than -1 t / +3 t	flyPad warning "INSERT UPLINK needs a TOW within -1 t / +3 t"; no INSERT UPLINK.
No TOW in the FMGC	flyPad warning "No TOW in the FMS (INIT B: ZFW and BLOCK)": fill INIT B first.
INSERT UPLINK	PERF TAKE OFF filled with the uplink.

Limitations and why

Limitation	Why	What to do
Only the thrust selected for the takeoff run is sent.	The airline sends what the crew asked for; the MCDU holds MAX and FLEX pages, the flyPad fills the one you use.	Select TOGA or FLEX before sending.
V1 = VR = V2 at light weights.	The FlyByWire A320 model (upstream).	Check against your own tools.
No real airline datalink.	The flyPad plays the airline.	-

How to use

1. On the MCDU, enter the flight plan with its departure runway and INIT B (ZFW, BLOCK).
2. flyPad > **Performance** > **Takeoff**: choose the **runway from the list**, fill the data, **Calculate**.
3. Choose TOGA or the FLEX temperature in TAKEOFF RUN, press **Send to FMS**.
4. MCDU PERF TAKE OFF > **<TO DATA** (6L) > **RECEIVED <TO DATA** (6L): check, then **INSERT UPLINK**.

How to test

#	Do this	Expected result
1	A32NX on the ground. MCDU: flight plan with a departure runway, INIT B ZFW and BLOCK. flyPad > Performance > Takeoff, runway from the list, conditions, TOW, Calculate.	RESULTS in green (FBW A320 model); THS shown when the T.O CG is below 27 %.
2	Press **Send to FMS**.	Toast "Takeoff data sent to the FMS: MCDU PERF TAKE OFF, UPLINK TO DATA, RECEIVED TO DATA"; MCDU scratchpad TAKEOFF DATA UPLINK.
3	MCDU PERF TAKE OFF > **<TO DATA** (6L) > **RECEIVED <TO DATA** (6L).	UPLINK MAX TO DATA (or FLX TO DATA with 4R): V1 / VR / V2 and their labels in column 11, not over the left labels.
4	Press **INSERT UPLINK**.	PERF TAKE OFF filled: V1, VR, V2, FLAPS/THS, FLEX TO TEMP (cleared for MAX), THR RED/ACC, ENG OUT ACC, TO SHIFT.
5	Change the INIT B block so that the TOW differs by more than +3 t, then Send to FMS again.	flyPad warning "INSERT UPLINK needs a TOW within -1 t / +3 t"; no INSERT UPLINK prompt.
6	Clear INIT B and send again.	flyPad warning "No TOW in the FMS (INIT B: ZFW and BLOCK)".
7	Choose a Manual runway on the flyPad and press Send to FMS.	Nothing is sent: the runway must come from the list.

4. A380X company datalink reply time

> **Aircraft:** A380X (the A32NX MCDU uplink stays instant) | **Where:** flyPad > Settings > Realism; MFD INIT, WIND, PERF T.O | **Status:** On the user's sim test list done ("datalink"); checked in the browser harness. Company winds with the FBW WIND page: covered by the sim test of the WIND page, PASS; unit tests CompanyReplyHold.spec and CompanyRequestTimeout.spec (plus the button rule specs).

What it is for

On the A380 the flight plan, the winds and the takeoff data come from the airline by datalink (FCOM DSC-22-FMS-10-40-90). A request takes time: the button shows **REQUEST PENDING...** until the answer, and **NO COMPANY REPLY**

appears when nothing comes within 4 minutes. FBW answered instantly; now the answer arrives after a reply time you choose.

The setting



Figure 26. flyPad > Settings > Realism: Company Datalink Reply Time, Real by default.

Setting	Answer to a request	Message sent without a request (e.g. SEND TO FMS)
Instant	immediately	immediately
Fast	5 to 15 s	1 to 3 s
Real (default)	60 to 120 s	5 to 15 s

Airbus gives no figure: these are typical airline values, below the 4 minute limit of the FCOM.

Requesting the flight plan

1. INIT > **CPNY F-PLN REQUEST** opens COMPANY F-PLN REQUEST (flight number, company route, free text).
2. Press **SEND F-PLN REQUEST**. The button becomes **REQUEST PENDING...**, here and on INIT; one request at a time.
3. After the reply time the SimBrief flight plan is delivered to the FMS, as before.
4. If SimBrief does not answer (no network, no flight plan), **NO COMPANY REPLY** appears after 4 minutes.

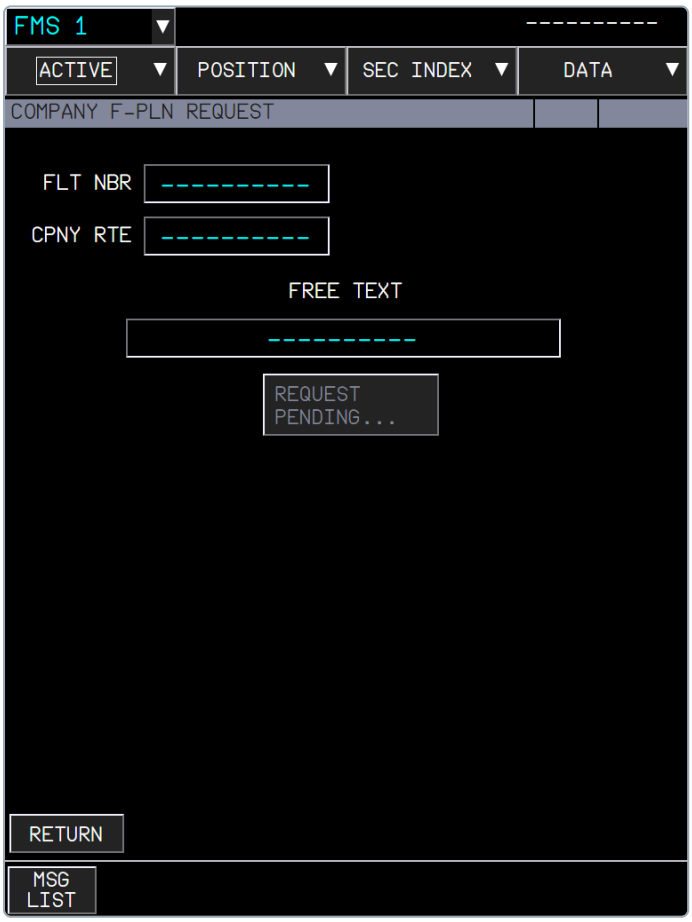


Figure 27. Request sent: REQUEST PENDING... (greyed until the answer).

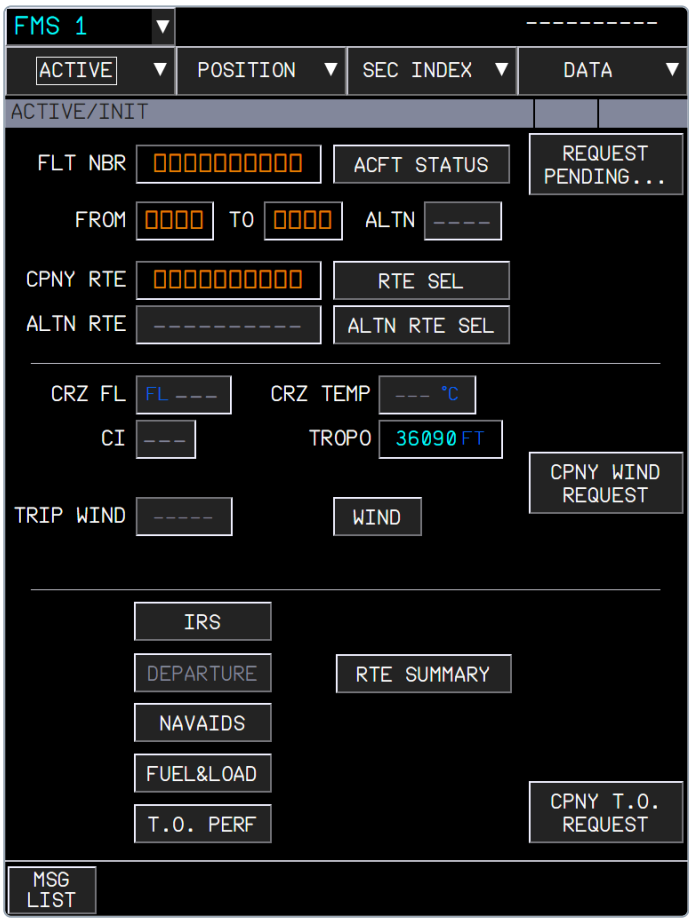


Figure 28. INIT shows the pending request too.

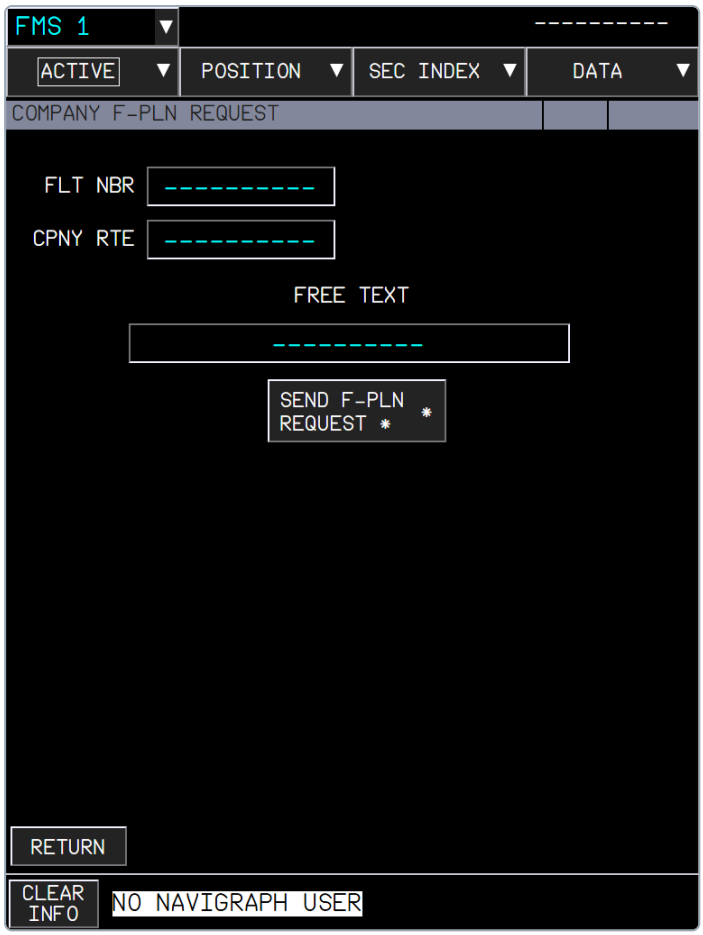


Figure 29. Without a Navigraph account or a SimBrief ID in the flyPad settings: NO NAVIGRAPH USER, no request.

Winds and takeoff data

- **CPNY WIND REQUEST** (INIT or WIND page, or SEND WIND REQUEST on the COMPANY WIND DATA REQUEST page): the winds are delivered after the reply time (see "Company winds" below).
- **Takeoff data** (PERF T.O > CPNY T.O REQUEST, see the A380 takeoff calculator above): the answer of the flyPad Send to FMS reaches the FMS not before the reply time of the request; without a request, after the transit time only.

What to expect

Situation	What you should see
Real, request sent	REQUEST PENDING... for one to two minutes, then the data.
Instant	The data at once, as before this change.
A second request while one is pending	Not possible: the button stays REQUEST PENDING.... (F-PLN request). For winds, the CPNY WIND REQUEST button of the plan that asked still opens the COMPANY WIND DATA REQUEST page, whose SEND WIND REQUEST button reads REQUEST PENDING.... and does nothing.
No answer	NO COMPANY REPLY after 4 minutes.
No Navigraph user / SimBrief ID	NO NAVIGRAPH USER; nothing is sent.

Limitations and why

Limitation	Why	What to do

The reply times are typical values.	Airbus publishes no figure for the airline answer.	Choose Fast or Instant if you prefer.
The A32NX MCDU uplink stays instant.	This change is for the A380 FMS; the A320 uplink has its own messages.	-

Company winds

The A380X WIND page and its company wind request come from FlyByWire master (merged into `core/a380x-mfd-fms`). FlyByWire's wind request goes through the ATSU and delivers the answer of SimBrief at once; two rules were put back on top of it, and a third closes a gap:

- **The reply time applies to the winds again.** The request time is recorded per flight plan and the answer is held until the reply time of the flyPad setting (Instant, Fast, Real) has passed; REQUEST PENDING... stays up meanwhile. Without this, only the company flight plan was delayed.
- **One wind request per flight plan, NO COMPANY REPLY after 4 minutes** (A380 FCOM DSC-22-FMS-10-40-90). The FMC remembers which flight plan asked: REQUEST PENDING... shows on the CPNY WIND REQUEST button and on the COMPANY WIND DATA REQUEST page of that plan only; on another plan the button is disabled (one company request at a time). If no answer comes within 4 minutes, NO COMPANY REPLY appears and a late answer is refused. Without this, a reload of the systems left REQUEST PENDING... up until a flight reload.
- **SEND F-PLN REQUEST is blocked while a wind request is pending** (the FMC dropped such a request silently).

How to use

1. flyPad > **Settings** > **Realism** > *Company Datalink Reply Time (F-PLN, Winds, T.O Data)*: Instant, Fast or Real (default).
2. Send the company requests as usual (CPNY F-PLN REQUEST, CPNY WIND REQUEST, CPNY T.O REQUEST) and wait for the answer while the button shows REQUEST PENDING....

How to test

#	Do this	Expected result
1	A380X on the ground, a SimBrief OFP available, Navigraph / SimBrief set in the flyPad. Set the reply time to **Real**.	Setting saved.
2	MFD INIT > **CPNY F-PLN REQUEST** > **SEND F-PLN REQUEST**.	REQUEST PENDING... (greyed) here and on INIT; the flight plan arrives 60 to 120 s later.
3	Press the request button again while it is pending.	Nothing happens: one request at a time.
4	Set **Instant** and request again (after clearing the plan).	The flight plan arrives at once.
5	INIT or WIND > **CPNY WIND REQUEST** with Real.	The winds arrive after the reply time.
6	Disconnect the network (or no OFP) and send a F-PLN request.	NO COMPANY REPLY after 4 minutes.
7	Reply time **Real**, a SimBrief flight loaded. WIND > **CPNY WIND REQUEST** (opens COMPANY WIND DATA REQUEST), SEND WIND REQUEST.	REQUEST PENDING... on the page and on the CPNY WIND button for the reply time (60 to 120 s), then the button reads RECEIVED CPNY WIND and COMPANY WIND DATA RECEIVED WAITING FOR INSERTION IN ACTIVE appears.
8	While the wind request is pending: MFD INIT > CPNY F-PLN REQUEST.	SEND F-PLN REQUEST is greyed out until the winds arrive.
9	While the wind request of the active plan is pending, open the WIND page of SEC 1.	The CPNY WIND REQUEST button there is disabled (one company request at a time).

10	Send a wind request and get no answer for 4 minutes (SimBrief unreachable).	NO COMPANY REPLY; an answer that arrives later is refused.
11	Remove the Navigraph user / SimBrief ID and press CPNY F-PLN REQUEST.	NO NAVIGRAPH USER; nothing is sent.

5. Landing calculator

> **Aircraft:** A380X and A32NX | **Where:** flyPad > Performance > Landing | **Status:** Tested in the sim by the user on both aircraft: OK, including the BTV exit advice on dry, wet and contaminated runways. Unit tests (mutation-tested).

What it is for

Before the descent, the crew checks that the aircraft can land on the planned runway: the landing distance for the runway condition and braking, the margin left, the approach speed VAPP and the go-around performance. On the real A380 this is the LDG PERF application of the OIS (FCOM PER-LND); on the A320 it is the QRH (in flight) and the FCOM tables (dispatch). This page does both, with the data of each aircraft.

- **IN-FLIGHT** (default): the actual landing distance for the braking mode you will use, as in the QRH / OIS, with the stop margin.
- **DISPATCH**: the regulatory (factored) required landing distance, to check a runway before departure.

The screen

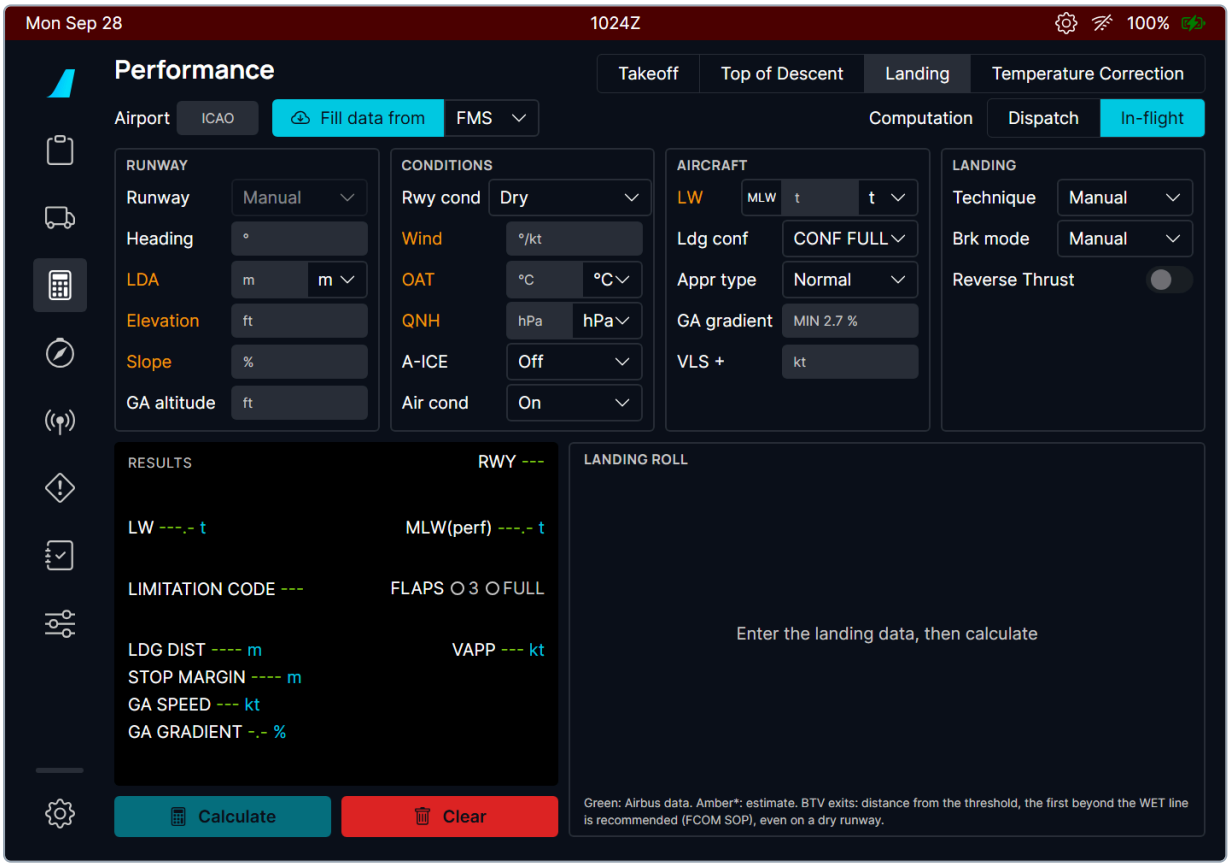


Figure 30. The A380 landing screen before any input. Amber labels are the inputs still missing for a calculation.

Part	Contents
Airport, Fill data from	The ICAO code; **Fill data from** imports the data from the **FMS** (A380 FMC / A320 MCDU), the **OFP** (SimBrief) or the **METAR**.
Computation	DISPATCH or IN-FLIGHT.

RUNWAY	Runway from the list (or Manual), heading, landing distance available (LDA), elevation, slope, and on the A380 the go-around altitude.
CONDITIONS	Runway condition, wind (direction/speed), OAT, QNH; on the A380 the anti-ice and air conditioning (they change the go-around gradient).
AIRCRAFT	Landing weight (MLW button = the maximum landing weight), landing configuration, and on the A380 the approach type, the required go-around gradient and the VLS increment.
LANDING	Manual landing or autoland, braking mode, reverse thrust, and on the A320 the overweight procedure.
RESULTS	The results in the OIS layout (below).
LANDING ROLL	A picture of the runway with the touchdown, the stop point of each braking mode, the stop margin, and on the A380 the BTV lines and the runway exits.

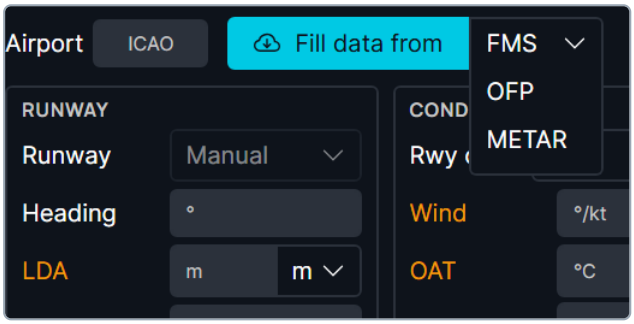


Figure 31. Fill data from: FMS (destination, runway, predicted landing weight and the PERF APPR QNH, temperature, wind and landing configuration), OFP or METAR.

A normal in-flight calculation (A380)

1. In flight, with the destination and the arrival runway in the flight plan and PERF APPR filled: choose **FMS** and press **Fill data from**. Otherwise type the airport, choose the runway and fill the conditions.
2. Check the landing weight (the FMS prediction) and the configuration.
3. Choose the braking mode you plan to use (manual, autobrake LO / 2 / 3 / HI, or BTV).
4. Press **Calculate**.

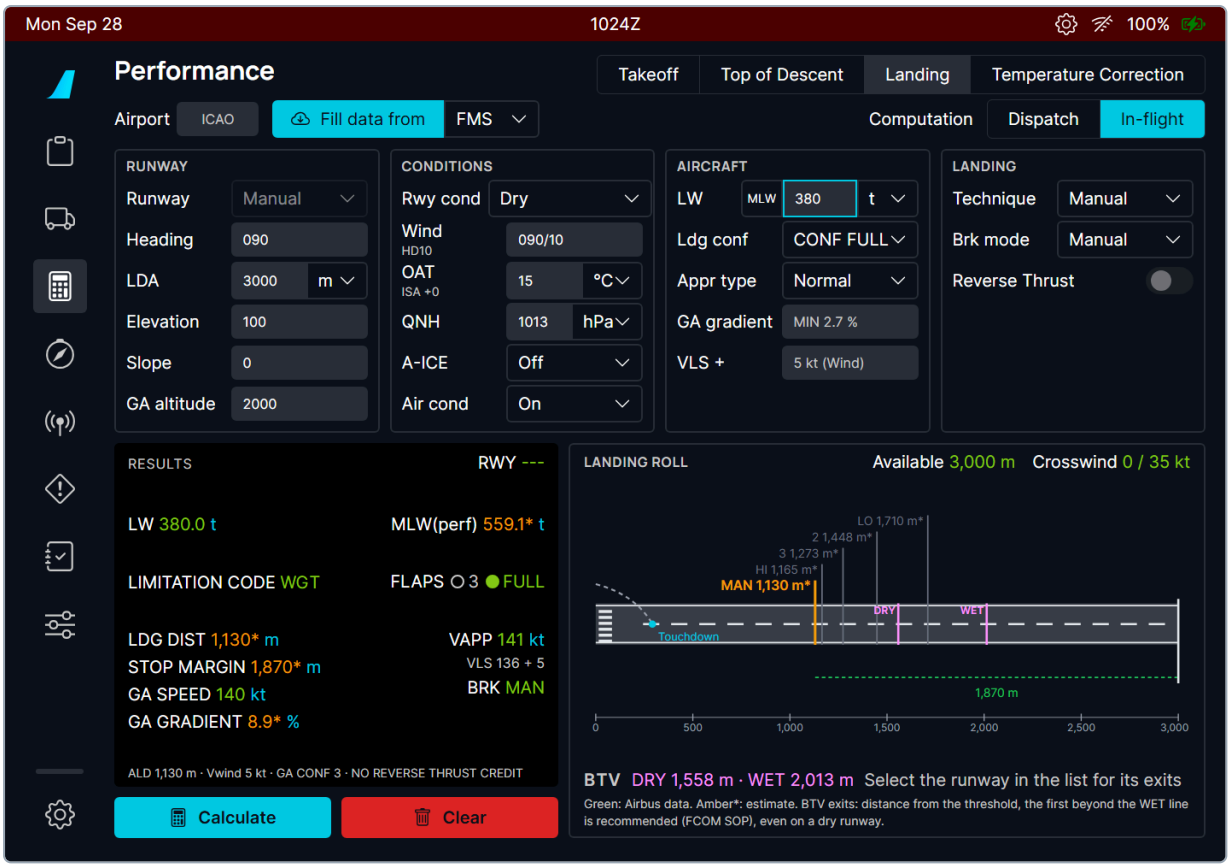


Figure 32. Dry runway, 3000 m, 380 t, CONF FULL, manual braking: landing distance 1130 m, stop margin 1870 m, VAPP 141 kt (VLS 136 + 5). The landing roll shows where each braking mode would stop.

Reading the results

Line	Meaning
LW	The landing weight used.
MLW(perf)	The **highest landing weight the runway and the go-around allow**: the lower of the weight whose landing distance just fits the LDA and the weight that still gives the required go-around gradient. It is a performance limit, not the structural maximum landing weight (395 t on the A380): it can be above it. Above the structural MLW the page shows OVERWEIGHT LANDING.
LIMITATION CODE	**WGT**: your weight is below MLW(perf), nothing limits you. **LDA** (red): the landing distance limits the weight. **ACG**: the go-around gradient limits it.
FLAPS	The landing configuration used (with AUTO CONF: the one the page chose).
LDG DIST / REG LDG DIST	In flight: the actual landing distance for the braking mode. Dispatch: the required (regulatory) distance.
STOP MARGIN	LDA minus the landing distance. Negative (red): the aircraft does not stop on the runway.
GA SPEED, GA GRADIENT	The go-around speed and the climb gradient of the go-around configuration.
VAPP	VLS of the landing configuration + the VLS increment (the wind increment, or what you enter).
Bottom line	The actual landing distance (ALD), the wind taken into account, the go-around configuration, and whether reverse thrust was credited.

Colours follow the flyPad takeoff calculator: **green** values come from Airbus data; values in **amber with a star (*)** are estimates of a model fitted on that data (see Limitations).

Braking modes

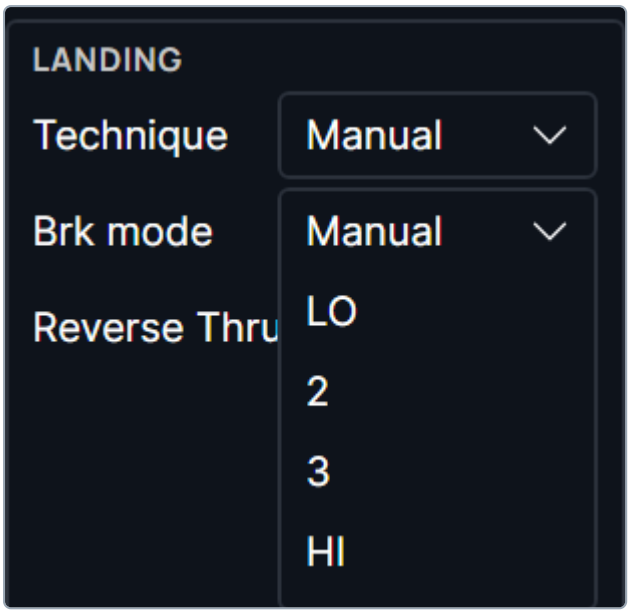


Figure 33. A380 braking modes: Manual, autobrake LO, 2, 3, HI (picture taken before the BTV mode was added: the list now ends with BTV, see "A380 landing calculator: BTV braking mode" below).

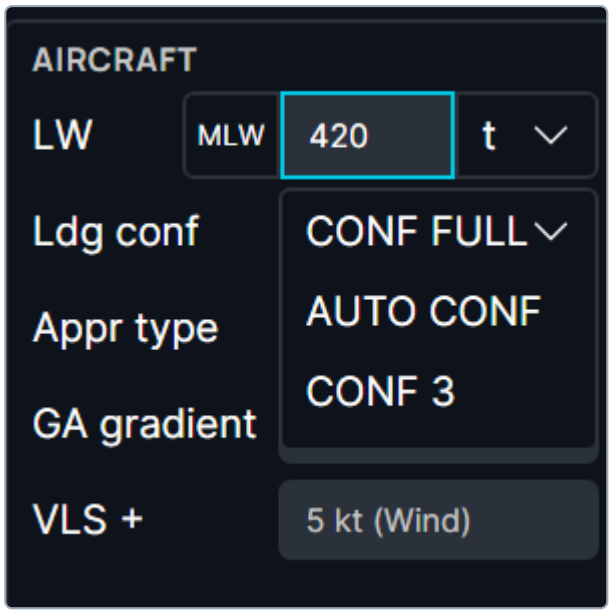


Figure 34. Landing configuration: CONF FULL, CONF 3 or AUTO CONF (the page chooses, as the OIS does).

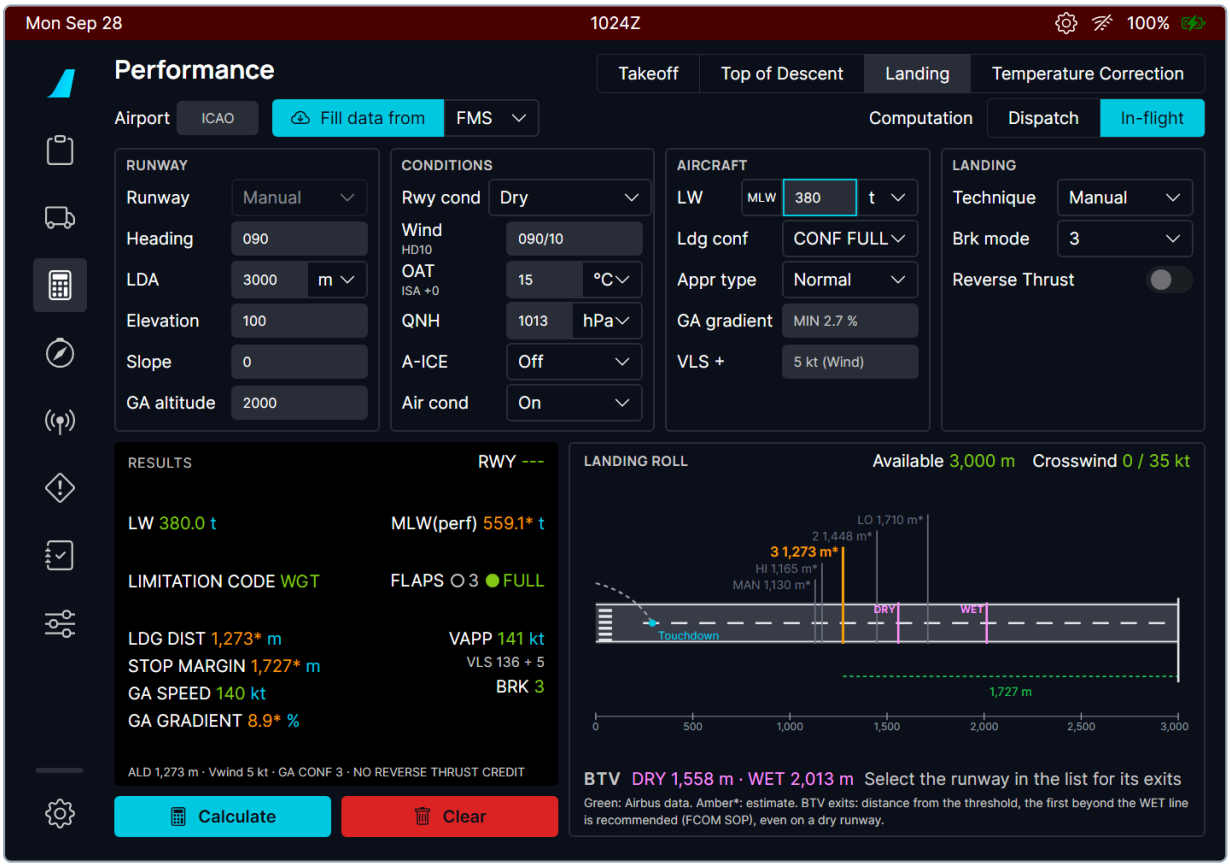


Figure 35. Same landing with autobrake 3: 1507 m. The stop point of the chosen mode is highlighted; the others stay visible for comparison.

AUTO CONF (A380, FCOM PER-LND): CONF FULL when its go-around in CONF 3 gives the required gradient, otherwise CONF 3 with a go-around in CONF 2; for an overweight landing, CONF 3 with a go-around in CONF 1+F.

Runway condition and reverse thrust

The runway condition follows the runway condition codes (RWYCC): dry, wet, water and slush 1/4" or 1/2", compacted snow, ice. The crosswind limit of the condition is shown at the top of the landing roll.

- **Reverse thrust** is credited only on a **contaminated** runway (FCOM: no credit on dry and wet runways); the result line says *REVERSE THRUST CREDIT* or *NO REVERSE THRUST CREDIT*.
- On the A380, **BTV is prohibited on a contaminated runway** (FCOM LIM-32-30): the BTV line says so and advises autobrake 3.
- With 1/2" of water or slush, a message reminds you that Airbus does not recommend these distances: a deeper contaminant gives a shorter distance in the data.

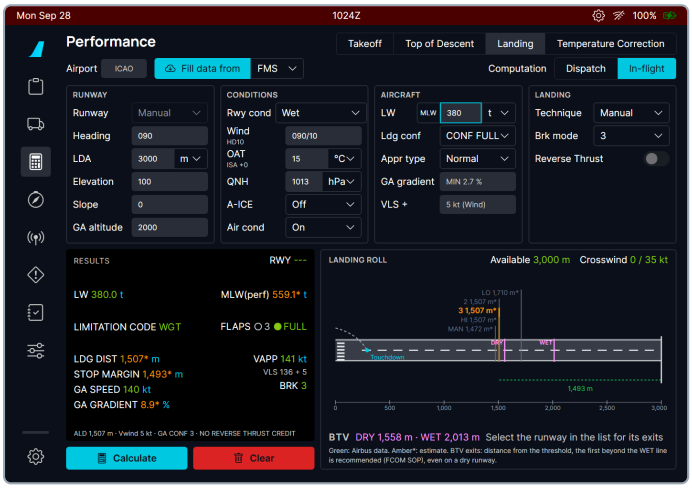


Figure 36. Wet runway, autobrake 3.

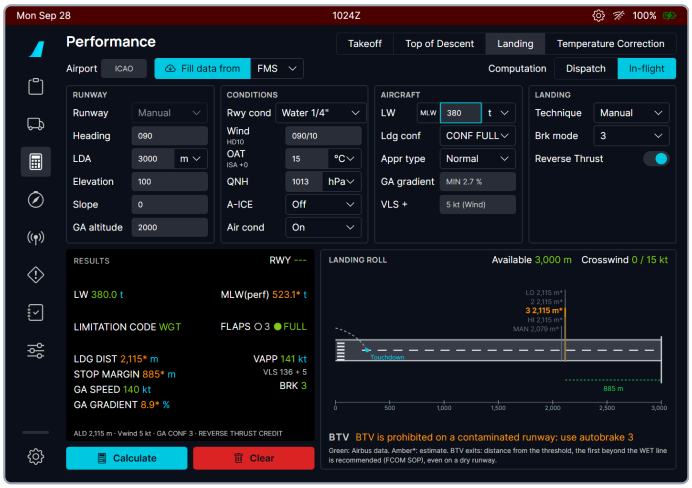


Figure 37. Water 1/4" with reverse thrust: credited, BTV prohibited, crosswind limit 15 kt.

BTV exits (A380)

On the A380, the page shows the **BTV DRY** and **WET** lines: where the aircraft stops with the BTV decelerations (FCOM DSC-32-10-30-20), from 400 m after the threshold plus 5 s of roll. When the runway comes from the list and a Navigraph account is linked, the runway exits of the airport map are drawn and rated:

Exit	Meaning
Recommended	The first exit beyond the WET line (the FCOM SOP choice, even on a dry runway).
Beyond WET	Achievable wet or dry.
Dry only	Between the DRY and WET lines: only on a dry runway.
Not achievable	Before the DRY line.

Without the runway list or the Navigraph data, the line says "*Select the runway in the list for its exits*". This is information for your briefing: the BTV selection itself is made on the OANS, as on the real aircraft.

Dispatch

DISPATCH gives the **required landing distance** used to check a runway before departure: the actual distance divided by 0.6 on a dry runway, ×1.15 on a wet one, and on a contaminated runway the higher of the wet value and ALD ×1.15 (A380 FCOM PER-LND). Braking is manual (the braking mode is greyed out). The orange bar is the required distance.

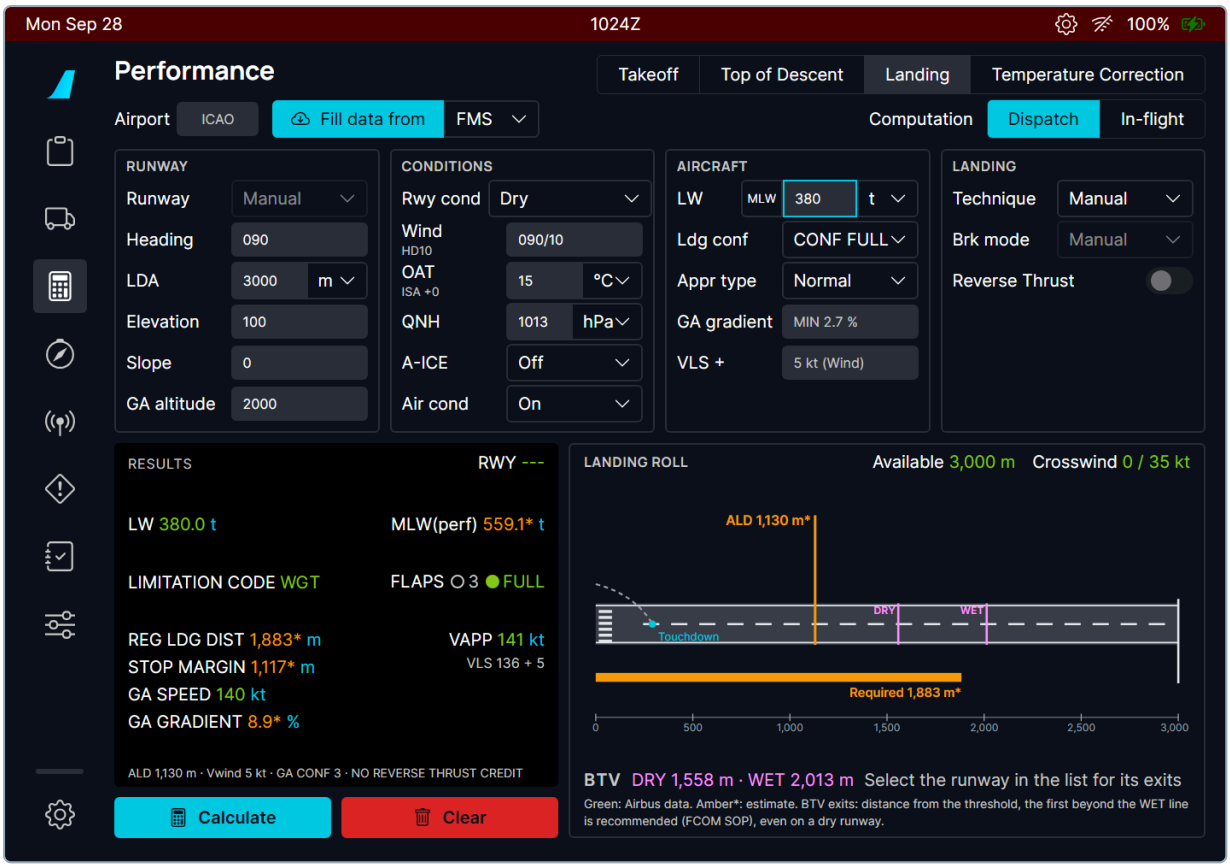


Figure 38. Dispatch on the same runway: required landing distance 1883 m (REG LDG DIST).

When the runway is too short or the aircraft too heavy

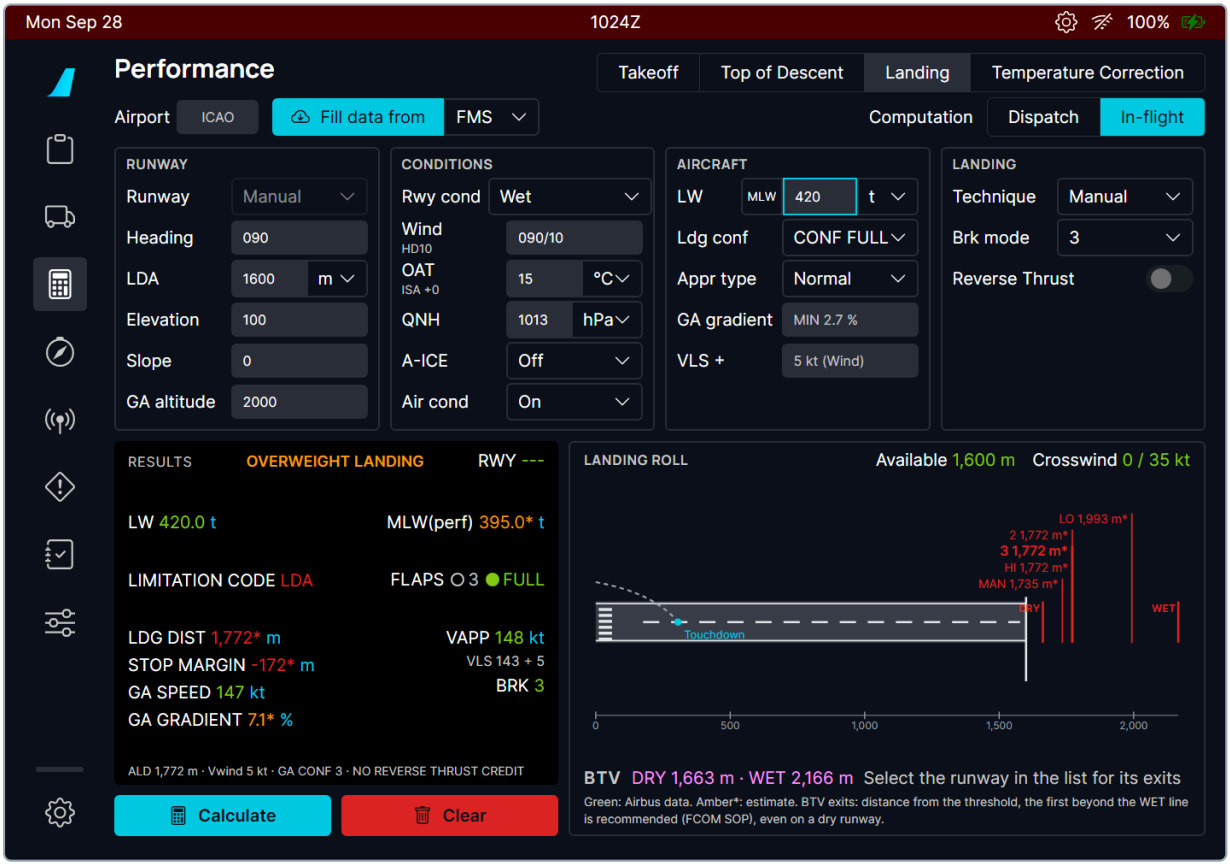


Figure 39. 1600 m wet runway at 420 t: OVERWEIGHT LANDING (above the 395 t MLW), limitation code LDA in red, negative stop margin (-172 m): the aircraft does not stop on this runway with autobrake 3.

The A320 version

The A320 uses the same screen with its own data: the **QRH in-flight landing distances** (shown ×1.15, the QRH margin) and the **FCOM dispatch tables** (PER-LDG-DIS-RLD) with the autoland correction and the wet minimum. Braking modes are Manual, LO and MED; the **Overweight Procedure** switch applies the QRH overweight landing corrections. There is no go-around section.

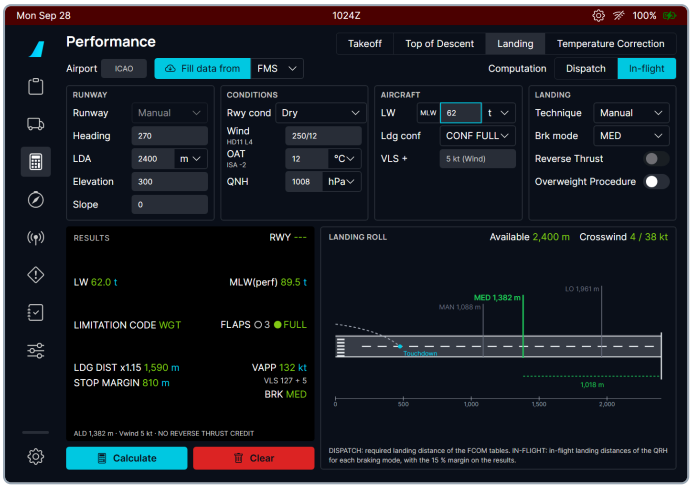


Figure 40. A320 in flight: 62 t, 2400 m dry, autobrake MED: 1382 m (1590 m with the 15 % margin).

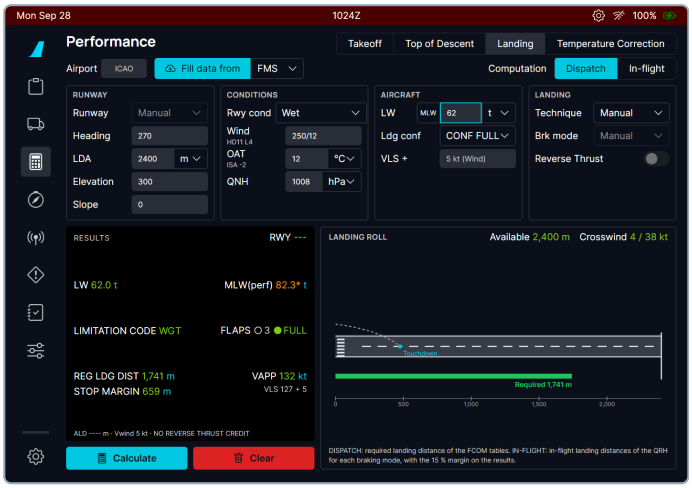


Figure 41. A320 dispatch on a wet runway: required 1741 m from the FCOM tables (green: Airbus data).

What to expect

Situation	What you should see
Inputs missing	Amber labels; the Calculate button stays dim until they are filled.
Fill data from FMS without a flight plan	Only what the FMS has is filled; complete the rest by hand.
Dry or wet runway, reverse thrust on	NO REVERSE THRUST CREDIT: the FCOM gives none there.
Contaminated runway	Longer distances, lower crosswind limit; A380: BTV prohibited.
Landing weight above the MLW	OVERWEIGHT LANDING in the RESULTS; A380 AUTO CONF lands in CONF 3.
LW above MLW(perf)	Limitation code LDA or ACG in red; the stop margin may be negative.
Change of an input after Calculate	Press Calculate again.

Limitations and why

Limitation	Why	What to do
Most A380 values are estimates (amber *).	The only public A380 landing data is the landing field length chart of the Airbus A380 Aircraft Characteristics (dry, manual, CONF FULL). The other cases use a model fitted on it (0.8 % error on the chart) and the FCOM rules.	Use the results for training and planning, not as certified data.
Autobrake and BTV distances are estimates.	They come from the autobrake decelerations of the FCOM, not from Airbus tables.	-
BTV exits need the runway from the list and a Navigraph account.	The exits come from the Navigraph airport map (the OANS data).	Choose the runway in the list; link Navigraph.
No BTV on the A320.	The A320 has no BTV.	-

The go-around gradient is calibrated on one FCOM example.	No A380 go-around tables are public.	-

How to use

1. In flight, with the destination, the arrival runway and PERF APPR (QNH, temperature, wind) in the FMS: flyPad > **Performance** > **Landing**, **Fill data from** FMS. Otherwise type the airport, choose the runway and enter the conditions.
2. Keep IN-FLIGHT (DISPATCH before departure), check the landing weight and the configuration, choose the braking mode and reverse thrust.
3. Press **Calculate**; read LDG DIST, STOP MARGIN, VAPP and the landing roll; on the A380 note the recommended BTV exit.

How to test

#	Do this	Expected result
1	A380X in flight (or on the ground with the destination entered), destination and arrival runway in the flight plan, PERF APPR filled. flyPad > Performance > Landing, Fill data from FMS.	Airport, runway, landing weight, QNH, temperature, wind and configuration filled.
2	Dry runway, manual braking, **Calculate**.	RESULTS: LW, MLW(perf), limitation code, FLAPS, LDG DIST, STOP MARGIN, GA SPEED / GRADIENT, VAPP; the landing roll with the stop point of each braking mode.
3	Choose autobrake 3 and Calculate.	A longer LDG DIST; the stop point of mode 3 highlighted.
4	With a Navigraph account linked and the runway from the list: look under the landing roll.	The BTV DRY and WET lines and the runway exits rated (recommended = first beyond WET; dry only; not achievable).
5	Set Water 1/4", reverse thrust on, Calculate.	REVERSE THRUST CREDIT; "BTV is prohibited on a contaminated runway: use autobrake 3"; lower crosswind limit at the top of the landing roll.
6	Dry runway with reverse thrust on.	NO REVERSE THRUST CREDIT.
7	Choose DISPATCH and Calculate.	REG LDG DIST (actual / 0.6 on dry); the braking mode is greyed.
8	IN-FLIGHT, landing weight 420 t on a short wet runway.	OVERWEIGHT LANDING, limitation code LDA in red, negative stop margin when the runway is too short.
9	A32NX: same page.	Braking modes Manual, LO, MED; Overweight Procedure switch; LDG DIST x1.15 in IN-FLIGHT; no go-around section.

6. A380 landing calculator: BTV braking mode

> **Aircraft:** A380X | **Where:** flyPad > Performance > Landing > LANDING > Brk mode | **Status:** Tested in the sim by the user: OK. Unit tests.

What it is for

The A380 can land with the Brake To Vacate (BTV) autobrake. The calculator gave the BTV DRY and WET lines and the exit advice, but BTV was not a braking mode. A380 FCOM DSC-32-20-220: the BTV landing distance is the BTV DRY line on a dry runway and the BTV WET line on a wet one (autoland plus autobrake distances until an exit is selected). **BTV** is now the last entry of the *Brk mode* list.

What you should see

Situation	What you should see
IN-FLIGHT, dry runway, Brk mode BTV	LDG DIST = the BTV DRY line of the landing roll; the BTV stop point is drawn with the other modes.
IN-FLIGHT, wet runway, BTV	LDG DIST = the BTV WET line.
Contaminated runway, BTV	Error "BTV: dry or wet runway only", no result (BTV is prohibited there, FCOM LIM-32-30). With another mode, the landing roll shows no BTV stop point.
DISPATCH	Braking is manual (the mode is greyed), as before.

Limitations and why

Limitation	Why	What to do
The BTV distance is an estimate (amber).	The BTV lines come from the BTV decelerations of the FCOM (DSC-32-10-30-20), not from Airbus tables.	Use it for planning only.
Selecting BTV here does not arm BTV in the cockpit.	On the aircraft the BTV runway and exit are selected on the OANS.	Select the exit on the OANS.

How to use

1. On the Landing page, in IN-FLIGHT, set *Brk mode* to **BTV** on a dry or wet runway.
2. Press **Calculate** and compare LDG DIST with the exits shown under the landing roll.

How to test

#	Do this	Expected result
1	A380X, flyPad > Performance > Landing, IN-FLIGHT, a runway from the list, dry, Brk mode **BTV**, Calculate.	LDG DIST equals the distance of the BTV DRY line; STOP MARGIN = LDA minus it.
2	Set the runway condition to Wet, Calculate.	LDG DIST equals the BTV WET line (longer).
3	Set Water 1/4" (or any contaminated condition) with BTV, Calculate.	Error "BTV: dry or wet runway only"; no result.
4	Keep the contaminated runway, choose autobrake 3, Calculate.	A result; the landing roll has no BTV stop point.

7. A380 landing calculator: 40 kt crosswind on dry and wet runways

> **Aircraft:** A380X | **Where:** flyPad > Performance > Landing | **Status:** Tested in the sim by the user on dry and wet runways: OK. Unit test.

What it is for

The landing calculator limited the crosswind to 35 kt on dry and wet runways (A380 FCOM PER-LND-CTA-CWD). The A380 FCOM revision for the EA engines (the GP7200 of the FBW A380X) increases the maximum certified crosswind at

landing from 35 kt to **40 kt**, gust included. The contaminated runway values are unchanged: 20 kt with slush or dry snow, 15 kt with standing water or compacted snow, 5 kt on ice. The 30 kt takeoff limit is unchanged too.

What you should see

Situation	What you should see
Dry or wet runway, crosswind up to 40 kt	The calculation runs without a question; the crosswind limit shown at the top of the landing roll is 40 kt.
Crosswind above 40 kt	A question "Crosswind above 40 kt": "The crosswind of N kt is above the maximum of 40 kt for this runway condition. Calculate anyway?"
Contaminated runway	The lower limit of the condition (20, 15 or 5 kt), as before.

How to use

1. Enter the wind as usual (or Fill data from FMS / METAR); the page computes the crosswind component.
2. If the question appears, answer it: the result is computed anyway when you accept.

How to test

#	Do this	Expected result
1	A380X, Landing page, a dry runway of heading 090, wind 180/38 kt, Calculate.	No crosswind question (before: question above 35 kt); limit 40 kt on the landing roll.
2	Wind 180/42 kt, Calculate.	Question "Crosswind above 40 kt ... Calculate anyway?".
3	Wet runway, wind 180/38 kt.	No question.
4	Water 1/4", wind 180/18 kt.	Question "Crosswind above 15 kt".

8. Descent calculator

> **Aircraft:** A380X and A32NX | **Where:** flyPad > Performance > Top of Descent | **Status:** Tested in the sim by the user: A32NX OK (after the fix of the last feature of this manual), A380X OK. Unit tests.

What it is for

The FMS computes the top of descent (T/D) for the managed descent. The crew still needs other answers: how long an emergency descent takes, where to start a descent at a given V/S, whether the aircraft is late on the profile after a shortcut, and a cross-check of the FMS T/D. The A380 OIS has an in-flight descent module for this (FCOM PER-IFT-DES); this page does the same on both aircraft, and replaces the old Top of Descent calculator.

It uses the **same flight and engine models as the FBW FMS predictions**, so its T/D agrees with the one drawn on the ND for the same inputs.

The screen

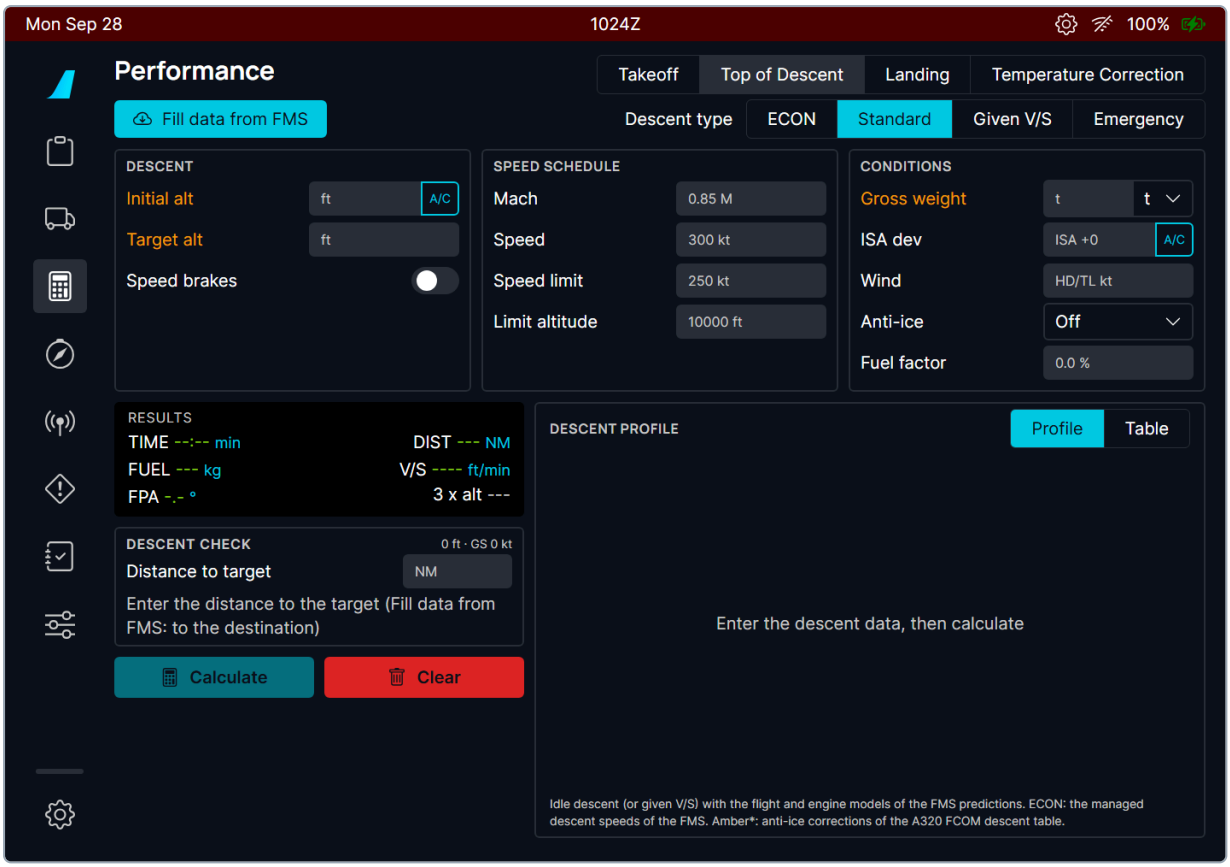


Figure 42. The descent calculator. Amber labels: inputs still needed.

Part	Contents
Fill data from FMS	Cruise level (or the current altitude when lower), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
Descent type	**ECON**: the managed descent speeds of the FMS. **Standard**: the FCOM speed schedule (A380 M.85 / 300 kt / 250 kt below FL100; A320 M.78 / 300 / 250). **Given V/S**: a constant vertical speed. **Emergency**: MMO / VMO with the speed brakes.
DESCENT	Initial altitude (**A/C**: the current one) and target altitude; speed brakes.
SPEED SCHEDULE	Mach, speed, speed limit and its altitude. Empty = the standard values shown in grey.
CONDITIONS	Gross weight, ISA deviation (**A/C**: from the current temperature), wind component (positive = headwind, negative = tailwind), anti-ice (off / engine / total), fuel factor.
RESULTS	Time, distance, fuel, mean V/S and flight path angle, and the "3 × altitude" rule of thumb for comparison; the crossover altitude and where the aircraft slows to the speed limit.
DESCENT CHECK	With the distance to the target: when to start down, or how late you are.
DESCENT PROFILE	Profile chart or table every 5000 ft.

Standard descent

1. Press **Fill data from FMS** in cruise with a flight plan, or type the initial altitude, the target altitude and the weight.
2. Choose the descent type and, if needed, the wind component and anti-ice.
3. Press **Calculate**.

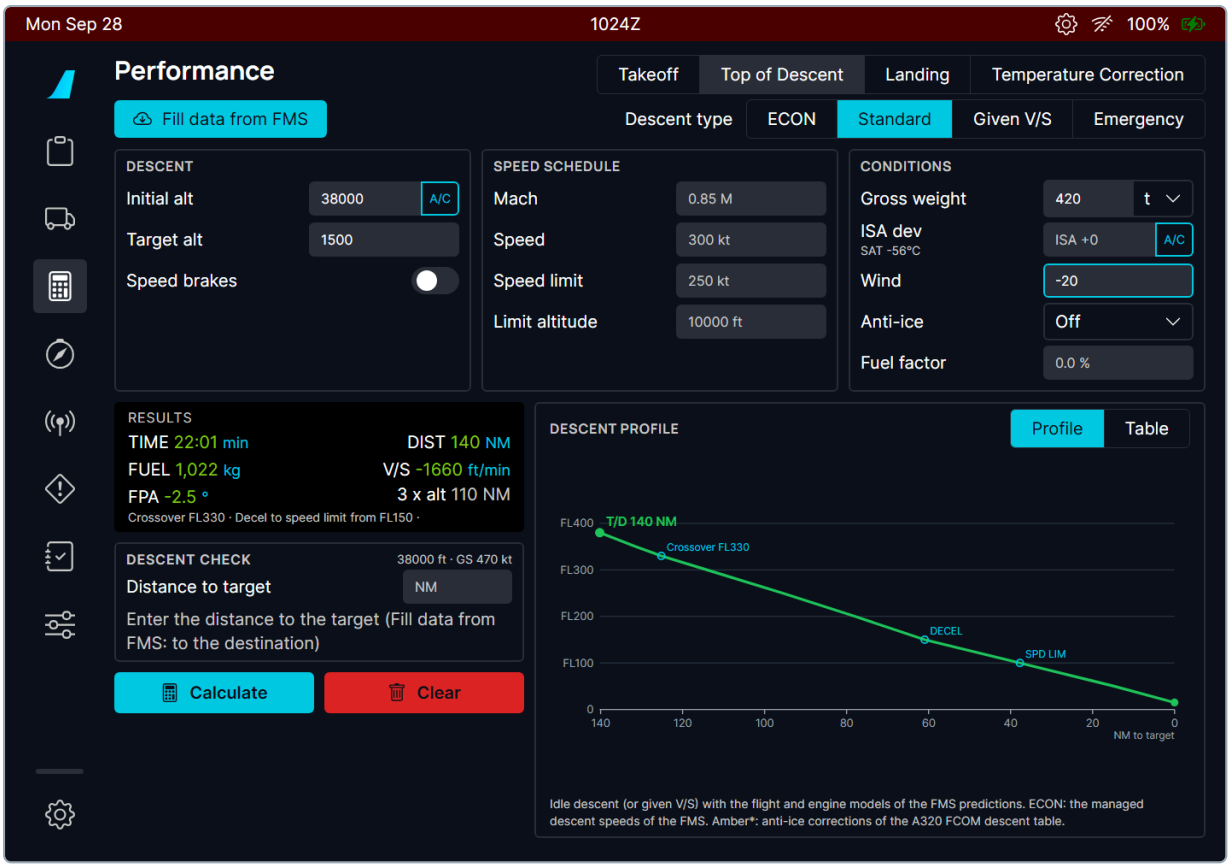


Figure 43. FL380 to 1500 ft, 420 t, 20 kt tailwind, standard speeds: 22 min, 140 NM, 1022 kg, -1660 ft/min on average (-2.5°). The chart shows the T/D, the crossover altitude (Mach to CAS), the deceleration (DECEL) and the speed limit (SPD LIM).

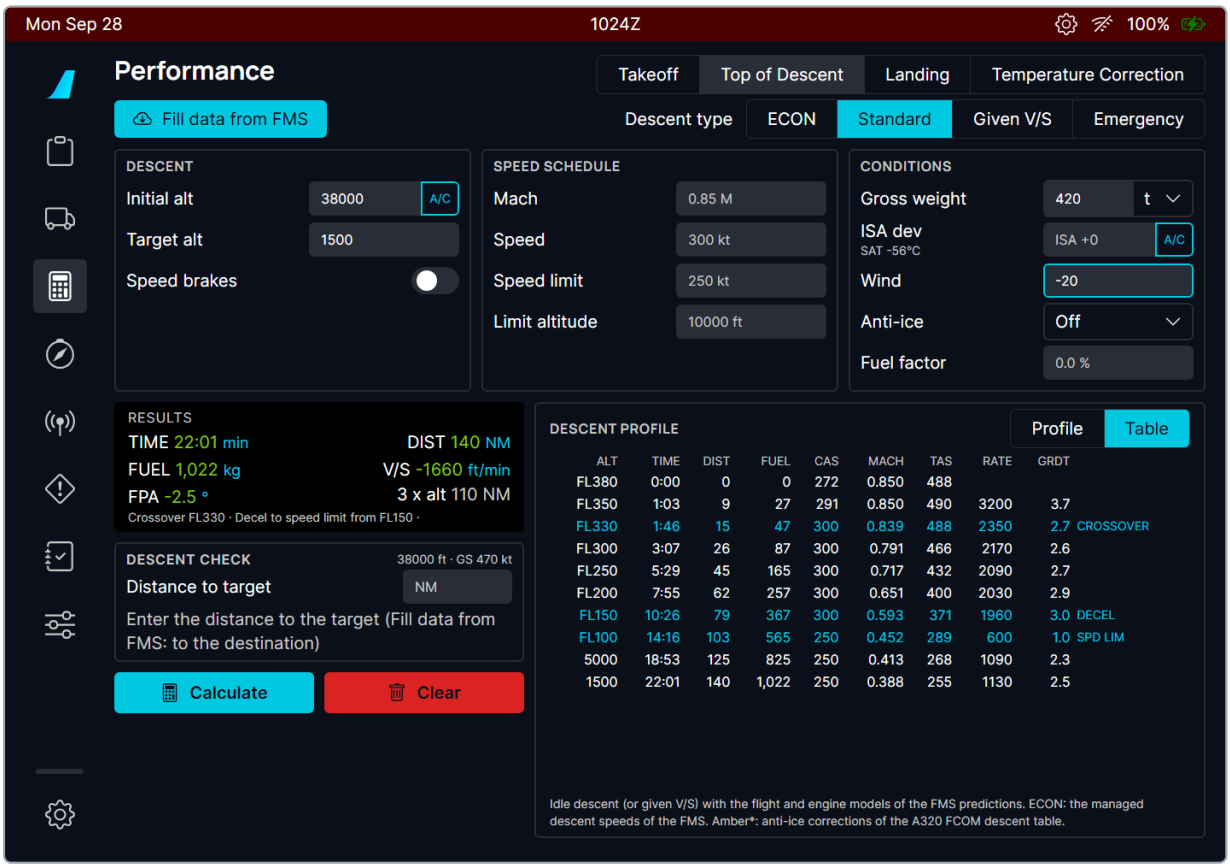


Figure 44. Table view: time, distance, fuel, CAS, Mach, TAS, RATE (rate of descent, ft/min) and GRDT (descent gradient, degrees) every 5000 ft, named and signed like the FCOM table results (PER-IFT-DES-DSR: both positive in descent); the crossover, DECEL and SPD LIM lines in blue. The RESULTS line gives V/S and FPA negative, as set on the FCU.

The deceleration to the speed limit is flown on the -1° idle path of the FMS descent strategy, which is why the profile flattens between DECEL and SPD LIM.

Descent check

Enter the **distance to the target** (Fill data from FMS enters the distance to the destination). The page reads the aircraft altitude and ground speed (shown top right of the box) and compares the aircraft with the profile. The aircraft is the triangle on the chart.

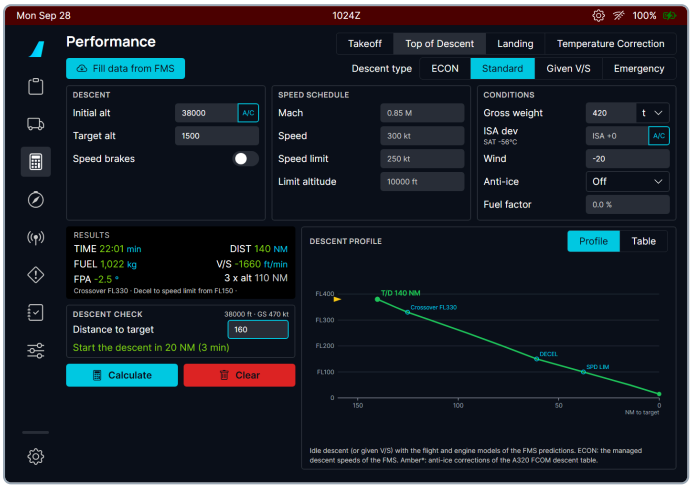


Figure 45. 160 NM to go, T/D at 140 NM: "Start the descent in 20 NM (3 min)".

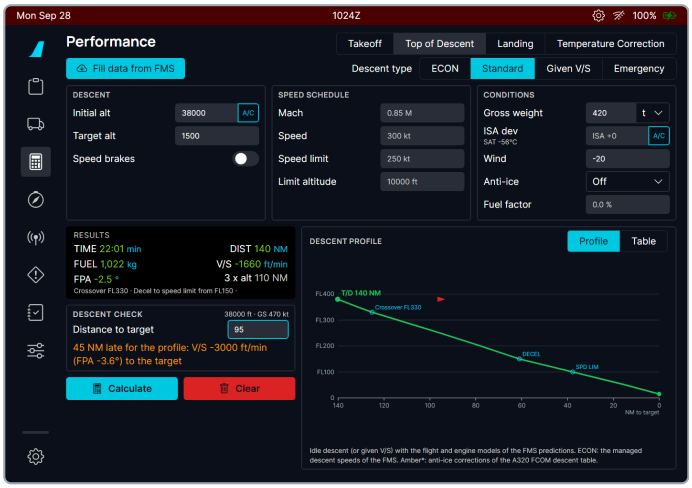


Figure 46. 95 NM to go at FL380: "45 NM late for the profile: V/S -3000 ft/min (FPA -3.6°) to the target".

> **TIP**
>
> After a shortcut, the late message gives the V/S (or FPA) that still makes the target, to set in V/S or FPA mode, or tells you to use the speed brakes.

Given V/S, emergency and ECON

Given V/S adds a V/S field. The page flies that V/S with the thrust it needs. Where the V/S is steeper than an idle descent can give, the page flies idle there and says so in amber (the aircraft cannot descend faster at idle without speed brakes).

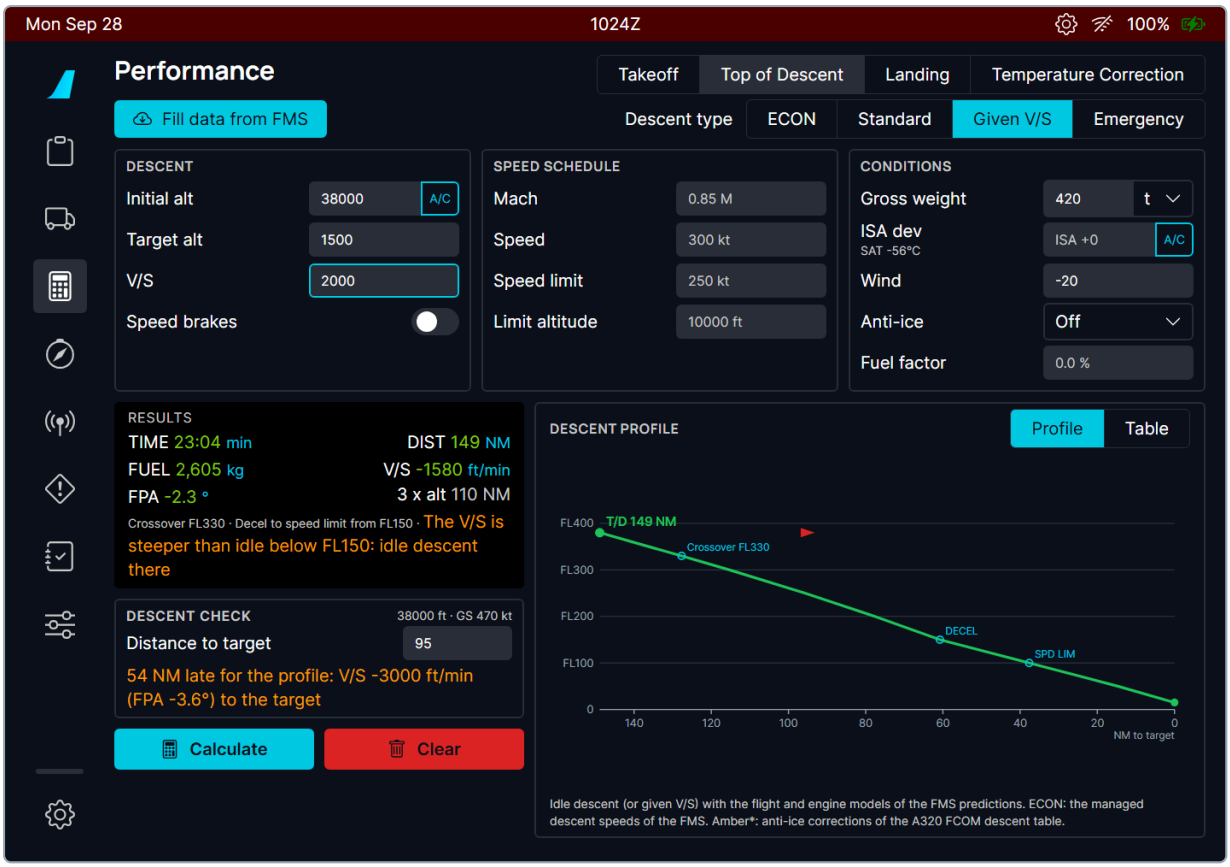


Figure 47. 2000 ft/min: shallower than idle at high level (149 NM, more fuel: 2605 kg), and steeper than idle below FL150, where the page flies idle.

Emergency uses MMO / VMO with the speed brakes extended (the schedule fields turn grey).

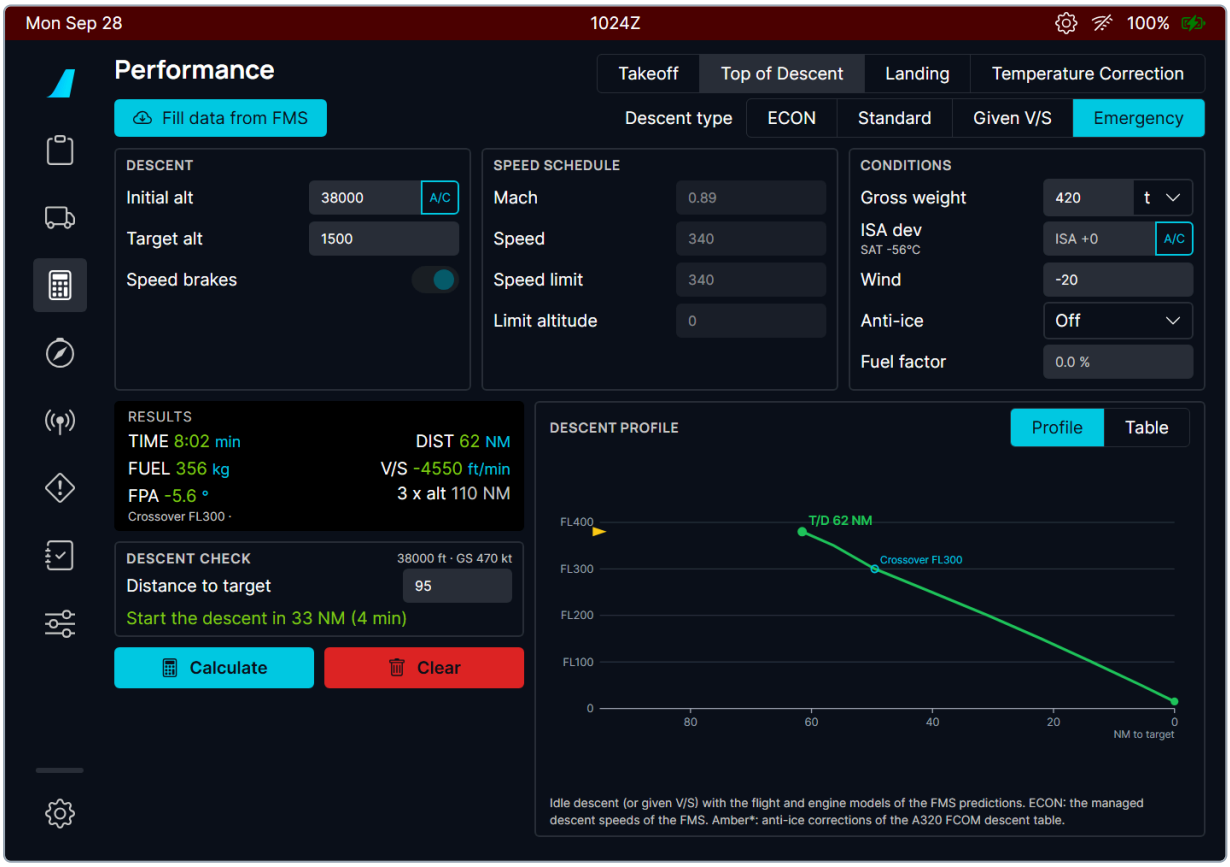


Figure 48. Emergency descent from FL380: 8 min, 62 NM, -4550 ft/min on average.

ECON needs the FMS: it asks the FMS for its managed descent speeds. Without a flight plan in the FMS, a message says so and the page stays on the previous type.



Figure 49. ECON without FMS data.

Anti-ice, fuel factor and estimates

Anti-ice (engine or total) raises the idle thrust and flattens the descent. Its effect uses the corrections of the A320 FCOM descent table (no A380 table is public): the values it changes turn **amber with a star**. The fuel factor (a percentage) scales the fuel burn, like the FMS PERF FACTOR.

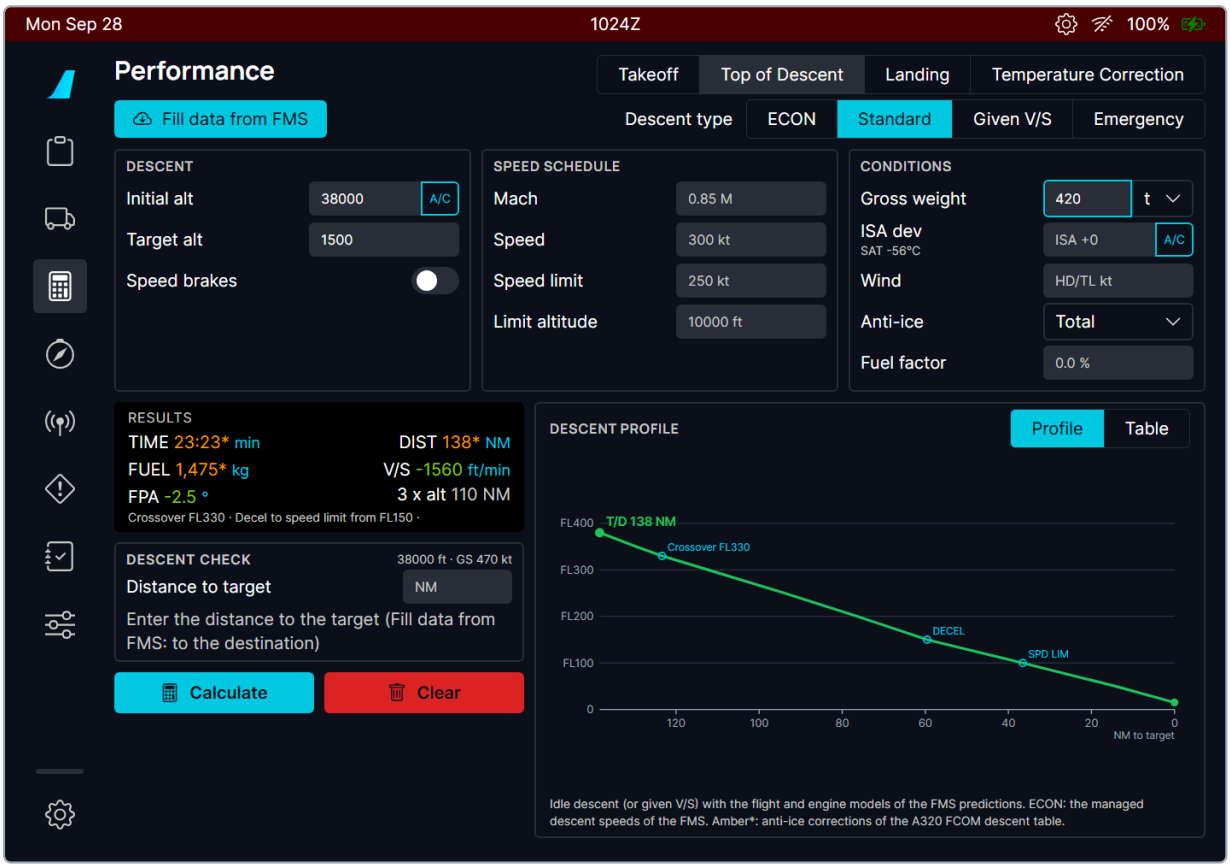


Figure 50. Total anti-ice: 138 NM, 23 min, 1475 kg, in amber (estimate).

What to expect

Situation	What you should see
Inputs missing	Amber labels; no results.
Any input changed	The results are cleared: press Calculate again.
Target above the initial altitude	Message "The target altitude must be below the initial altitude".
Initial altitude above the aircraft ceiling, or weight out of range	A message gives the limit (maximum altitude, or the weight range from the empty weight to the MTOW).
Given V/S steeper than idle	Amber line: idle descent below the altitude it names.
ECON without FMS flight plan	"No FMS data: enter a flight plan in the FMS".
Distance to target larger than the descent	Green "Start the descent in X NM (Y min)".

Distance shorter than the descent	Amber "N NM late for the profile" with the V/S and FPA needed.

Limitations and why

Limitation	Why	What to do
The descent is a still-air profile plus a mean wind.	The wind is one component for the whole descent, as in the FCOM and OIS; the FMS uses the wind of each altitude.	Use the mean wind of the descent.
About 8 % longer than the A320 FCOM table.	The FCOM table is for the CFM56 A320; the FBW A320neo engine model idles differently.	Compare with the FMS T/D, which uses the same model.
Anti-ice values are estimates.	Only the A320 FCOM has anti-ice descent corrections.	-
No descent path constraints or approach.	The page computes a clean descent to one target altitude, not the FMS path with its constraints.	Use it as a cross-check of the FMS.

Descent integrated at the printed speed

The A32NX descent calculator (fmgc FmsDescentPerformanceCalculator) integrated the descent at the speed limit whenever no deceleration step was needed, while the table printed the schedule speed: a descent to FL100 was integrated at 250 kt all the way. One rule now gives the speed at each altitude for both the integration and the table: the schedule speed above the limit altitude or until the deceleration, the lower of the two speeds at and below it. A descent that ends at the limit altitude decelerates before reaching it.

How to use

1. In cruise with a flight plan: flyPad > **Performance** > **Top of Descent**, **Fill data from FMS** (or type the initial and target altitudes and the weight).
2. Choose the descent type (ECON, Standard, Given V/S, Emergency), the wind component and anti-ice.
3. Press **Calculate**; read TIME, DIST, FUEL, V/S and FPA, and the DESCENT CHECK line (start in X NM, or late with the V/S needed).

How to test

#	Do this	Expected result
1	A380X in cruise (e.g. FL380) with a flight plan to a destination. Top of Descent page, **Fill data from FMS**.	Initial altitude, target (destination elevation + 1500 ft), gross weight, speed limit, ECON speeds and the distance to the destination filled.
2	Descent type Standard, **Calculate**.	RESULTS TIME / DIST / FUEL / V/S / FPA and 3 x alt; the profile with T/D, crossover, DECEL and SPD LIM.
3	Switch to the Table view.	Rows every 5000 ft with RATE and GRDT positive; the RESULTS line keeps V/S and FPA negative.
4	Compare the T/D distance with the FMS T/D on the ND.	About the same (same models).
5	Enter a distance to target longer than DIST.	Green "Start the descent in X NM".
6	Enter a distance shorter than DIST.	Amber "N NM late for the profile: V/S ... (FPA ...) to the target".
7	Given V/S 2000 ft/min, Calculate.	Amber line "The V/S is steeper than idle below ...: idle descent there" where it applies.
8	Emergency, Calculate.	MMO / VMO with speed brakes; schedule fields grey.

9	Repeat on the A32NX.	Same page and behaviour with the A320 speeds (M.78 / 300 / 250).

9. Descent calculator: T/D distance slider

> **Aircraft:** A380X and A32NX | **Where:** Top of Descent page, under the profile | **Status:** No separate sim-test record of the slider (the descent page itself was tested OK, see above); checked in the browser harness.

What it is for

After a calculation, a quick answer to "what if I start down earlier (vectors) or later (a shortcut)?" without a new calculation: the V/S and path angle the moved descent needs, against the calculated ones.

What if the descent starts earlier or later?

After a calculation, the **T/D distance** slider under the profile moves the start of the descent: to the left farther from the target (earlier, e.g. vectors), to the right closer (later, e.g. a shortcut), up to 60 NM each way. The chart draws the moved top of descent dashed, straight to the target, and the line under the slider gives the V/S and the path angle it needs, to compare with the calculated ones.

- **Later**: steeper than the idle descent: the figures turn amber, "speed brakes or a higher speed needed" (red when the calculation already used the speed brakes: start earlier).
- **Earlier**: shallower than the idle descent: "thrust needed, more fuel".
- The figures keep the mean ground speed of the calculated descent: an estimate for the decision, not a new calculation.

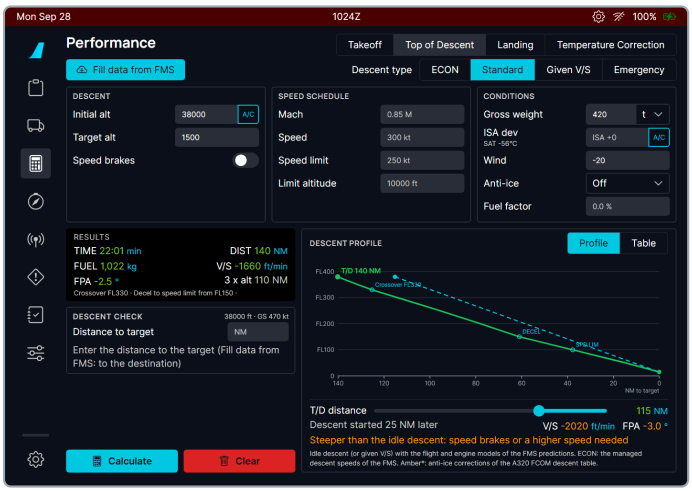


Figure 51. Slider 25 NM later: 115 NM, -2020 ft/min, -3.0°: steeper than idle.

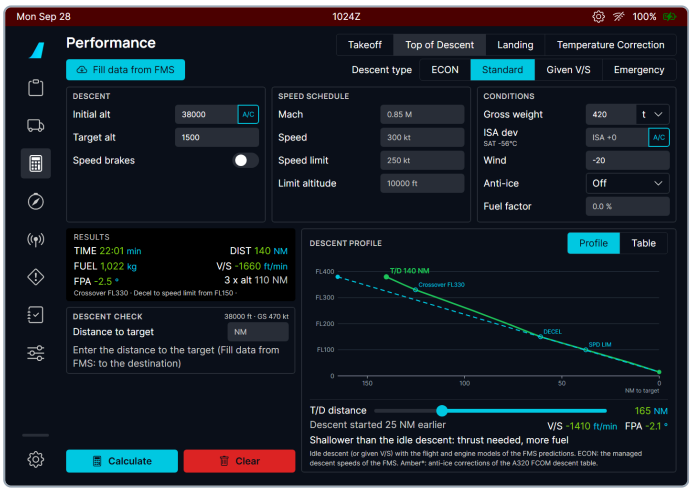


Figure 52. Slider 25 NM earlier: 165 NM, -1410 ft/min, -2.1°: shallower.

The **Calculate** and **Clear** buttons are at the bottom of the page, where they are on the takeoff and landing pages.

How to use

1. Calculate a descent.
2. Move the **T/D distance** slider under the profile: left = earlier (farther from the target), right = later.
3. Read the line under the slider: the V/S and FPA needed, and whether speed brakes or thrust are needed.

How to test

#	Do this	Expected result

1	Calculate a standard descent from FL380 (e.g. 420 t, 20 kt tailwind).	Slider centred: "Descent as calculated".
2	Move the slider 25 NM to the right.	"Descent started 25 NM later", steeper V/S and FPA in amber, "Steeper than the idle descent: speed brakes or a higher speed needed"; dashed line from the moved T/D to the target.
3	Move it 25 NM to the left.	"Descent started 25 NM earlier", shallower values, "Shallower than the idle descent: thrust needed, more fuel".
4	Emergency type (speed brakes in), slider to the right.	Red "Steeper than the descent with speed brakes: start the descent earlier".
5	Move the slider left, back to the T/D, right, and back again.	One "T/D xxx NM" label all the time; the profile keeps its size while the note line under the slider comes and goes.
6	Look at the buttons.	Calculate and Clear at the bottom of the page.

10. Descent calculator: target waypoint from the flight plan

> **Aircraft:** A380X and A32NX | **Where:** Top of Descent page > DESCENT > Target wpt | **Status:** Tested in the sim by the user: OK. Unit tests.

What it is for

The target of a descent is often a waypoint with an altitude constraint (the first constraint of the STAR, a crossing restriction from ATC), not the destination. The FMS (A380 FMC, A32NX MCDU) now sends the waypoints ahead of the aircraft, up to the missed approach, with their altitude constraints and their distance along the route. The **Target wpt** list of the flyPad picks one as the target.

What you see

- The list shows each waypoint with its constraint written as on the F-PLN: `9000` (at), `+9000` (at or above), `-9000` (at or below), `11000/9000` (window). A dash = no target waypoint (the destination).
- Picking a waypoint sets the **target altitude** from its constraint (the upper altitude of a window, to the nearest 100 ft; a waypoint without a constraint keeps the target altitude you entered) and the **distance to target** along the flight plan.
- The distance follows the aircraft: it is refreshed every 5 s. When the waypoint is sequenced, the target waypoint is cleared.
- Typing a distance yourself clears the target waypoint (the distance is no longer the waypoint's).

Limitations and why

Limitation	Why	What to do
The waypoints need the FMS.	They come from the active flight plan.	Without a flight plan the list holds only the dash.
Climb constraints of the SID are not used as targets.	An "at or above altitude 2" constraint is a climb constraint.	-

How to use

1. In cruise, with a STAR in the flight plan: Top of Descent page, **Fill data from FMS**.

2. Open **Target wpt** and pick the waypoint with the constraint you must meet.
3. Press **Calculate** and watch the DESCENT CHECK line as you fly.

How to test

#	Do this	Expected result
1	A380X or A32NX in cruise with a flight plan that has a STAR with altitude constraints. Top of Descent page.	The Target wpt list shows the waypoints ahead with their constraints (e.g. `ABCDE +11000`).
2	Pick a waypoint with an "at" constraint.	Target alt = the constraint; Distance to target = its distance along the flight plan (compare with the F-PLN).
3	Pick a waypoint with a window constraint.	Target alt = the upper altitude of the window.
4	Calculate and wait 10 s.	The distance to target decreases as the aircraft flies (refresh every 5 s).
5	Type a distance in the Distance to target field.	The Target wpt list goes back to the dash.

11. Descent calculator with an FMS that sends no waypoints

> **Aircraft:** A32NX (any aircraft whose FMS sends no waypoint list) | **Where:** Top of Descent page | **Status:** User-confirmed in the sim on the A32NX: no more crash. Unit tests.

What it is for

An FMS built before the waypoints were added to the descent data (for example an older A32NX MCDU) answers without a waypoint list. The page then crashed with the flyPad error screen ("fmsWaypoints.map is not a function"). The descent data now always carries a waypoint list, empty when the FMS sends none.

What you should see

Situation	What you should see
FMS without waypoints	The page opens and Fill data from FMS works; the Target wpt list holds only the dash.
Current FMS	The waypoints as in the previous feature.

How to use

1. Nothing to do: open the Top of Descent page as usual.
2. After an update, install the flyPad and the FMS (MCDU / MFD) together: they exchange the descent data.

How to test

#	Do this	Expected result
1	A32NX in flight with a flight plan. flyPad > Performance > Top of Descent.	The page opens (no error screen).
2	Press **Fill data from FMS**, then Calculate.	Data filled, results shown, no crash.
3	Delete the flight plan on the MCDU, open the page again and press Fill data from FMS.	No crash; a message says what the FMS lacks (e.g. "No FMS data: enter a flight plan in the FMS"); the Target wpt list holds only the dash.

12. Shared landing calculators above the tailwind limit

> **Aircraft:** A380X and A32NX (shared calculators; the flyPad page of this branch is unchanged) | **Where:** Shared landing calculators (used by the A380X OIT LDG PERF of core/a380x-ois); flyPad > Performance > Landing | **Status:** Not sim-tested on the core branch. On develop (6d0c08e45, with its flyPad part): checked in the browser harness on both aircraft, no sim test recorded. Calculator unit tests.

What it is for

The shared A380 and A320 landing calculators no longer stop with an error when the tailwind is above the limit of the aircraft data (A380 10 kt, A320 15 kt, compared in whole knots). The distances, the stop margin and the MLW(PERF) continue the tailwind correction of the data, and the result is flagged as extrapolated. This is so that the A380X OIT LDG PERF of `core/a380x-ois` calculates like develop. The flyPad landing page of this branch reads the flag and keeps stopping at the limit with its message, as before.

Limitations and why

Limitation	Why	What to do
The flyPad page shows no figures above the tailwind limit.	The extrapolated display, the red wind banner and the Wind check tab need the redesigned flyPad page kit and are on `core/efb-redesign`.	Use core/efb-redesign, or the OIT LDG PERF of core/a380x-ois.
A tailwind less than half a knot above the limit is no longer refused.	The limit is now compared in whole knots (10.4 kt counts as 10 kt).	-

How to use

1. flyPad > Performance > Landing: as before; above the tailwind limit the page gives its message and no result.

How to test

#	Do this	Expected result
1	A380X, flyPad > Performance > Landing: a runway and a wind giving a 15 kt tailwind. Calculate.	No result; the message "The tailwind is above the maximum of 10 kt" (as before).
2	Change the wind to a 10 kt tailwind. Calculate.	Results shown.
3	A32NX: same page with a 16 kt tailwind.	The message with 15 kt; no result.